

The magnitude and time course of pre-saccadic foveal prediction depend on the conspicuity of the saccade target

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eLife Assessment

This study reports **important** findings about pre-saccadic foveal prediction and the extent to which it is influenced by the visibility of the saccade target relative to its background. The results and research methodology are **solid**, although the neural substrates of oscillatory pre-saccadic enhancement for high-opacity targets remain unclear. This work should be of broad interest to visual neuroscientists, as well as those aiming to understand perception in the context of eye movements and modeling of visually guided actions.

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Abstract

We previously demonstrated that during the preparation of a large-scale saccadic eye movement, defining features of the eye movement target (i.e., its orientation) are anticipated in pre-saccadic foveal vision (Kroell & Rolfs, 2022). In this Research Advance, we show that the conspicuity of orientation information at the saccade target location influences the magnitude and time course of foveal enhancement. As in our previous study, observers prepared a saccade to a peripheral orientation signal (the target) while monitoring the appearance of a second orientation signal (the probe) in their pre-saccadic center of gaze. The probe appeared in 50% of trials and either had the same orientation as the target (congruent) or a different orientation (incongruent). In the current study, we manipulated the opacity of the target against the 1/f background noise in four logarithmic steps from 25% to 90%. An increase in opacity translated to an increase in luminance contrast and the signal-to-noise ratio of orientation information within the target region. We made three main observations: First, foveal Hit Rates for target-congruent *and* incongruent probes decreased as target opacity increased, presumably since attention was increasingly drawn to the target the more salient it became. Crucially, foveal enhancement defined as the difference between congruent and incongruent Hit Rates increased with opacity. Second, the time course of foveal enhancement appeared to show an oscillatory pattern that was particularly pronounced at

higher target opacities. Interestingly, the temporal frequency of these oscillations corresponded to the frequency range typically associated with neural feedback signaling. Third, unlike the difference in Hit Rates, the difference in False Alarm Rates did not vary with opacity. Instead, reverse correlations suggest that at higher target opacities, False Alarms were increasingly triggered by signal, that is, by incidental orientation information in the foveal noise. Beyond providing new mechanistic insights into active foveal processing, these findings are relevant for researchers planning to adapt our paradigm to study related questions. Presenting the saccade target at a high signal-to-noise ratio appears beneficial as congruency effects, especially when time-resolved, are most robustly detectable.

Introduction

We previously demonstrated that during the preparation of a saccadic eye movement to a peripheral visual field location, defining features of the eye movement target are predictively enhanced in pre-saccadic foveal vision (Kroell & Rolfs, 2022 [DOI](#)). In this Research Advance, we manipulated the conspicuity of the saccade target in a parametric fashion and, based on a large number of trials, describe the influence of this manipulation on the magnitude and temporal development of foveal prediction.

In our original paradigm, we presented a rapid sequence of 1/f noise images filling the entire screen (see Hanning & Deubel, 2022a [DOI](#); 2022b [DOI](#)). Human observers maintained fixation in the screen center until an orientation signal, generated by filtering the luminance content at the desired display location to either a -45° or 45° tilt, appeared 10 degrees of visual angle (dva) to the left or the right of the screen center. Observers executed an eye movement to the peripheral orientation signal (i.e., the target) as soon as they detected it. During saccade preparation, a second orientation signal (the foveal probe) could appear in observers' current, pre-saccadic center of gaze on 50% of trials. If presented, the probe either had the same orientation as the target (congruent) or a different orientation (incongruent). After executing the saccade, observers reported if they had perceived the probe in the screen center before moving their eyes and, if so, which orientation they had perceived.

Starting 175 ms before saccade onset, observers' Hit Rates (HRs) for foveal probes with target-congruent orientation significantly exceeded incongruent HRs. This congruency effect was spatially confined to the center of gaze and its immediate vicinity and more pronounced during saccade preparation than during passive fixation. Based on these findings, we proposed that feedback connections from higher-order visual areas relay relevant saccade target features to foveal retinotopic cortex (see also Williams et al., 2008 [DOI](#); Chambers, Allen, Maizey, & Williams, 2013 [DOI](#); Fan, Wang, Shao, Kersten, & He, 2016 [DOI](#); Weldon, Rich, Woolgar, & Williams, 2016 [DOI](#); Yu & Shim, 2016 [DOI](#); Weldon, Woolgar, Rich & Williams, 2020 [DOI](#)). Feed-back orientation signals subsequently combine with feedforward foveal input to early visual cortex and facilitate the detection of target-congruent feature information in the center of gaze.

In all previous experiments, we presented the saccade target at an opacity of 60% against the background noise. Importantly, the choice of a certain target opacity in our paradigm dictates the luminance contrast as well as the signal-to-noise ratio (SNR) of orientation information within the saccade target region. We had chosen a medium target opacity in previous experiments since we presumed that feature information may need to be relevant for the movement task to be predicted foveally. In other words, while high target opacities would likely have generated a salient and immediate target pop-out, continuous peripheral sampling of orientation information (see Ludwig, Davies, & Eckstein, 2013) may have been necessary to program the eye movement at a medium opacity. On the other hand, a highly conspicuous saccade target may not only facilitate the oculomotor task and free up resources for perceptual detection but could benefit foveal prediction more directly: The higher the SNR of peripheral orientation information, the less noisy the foveally

predicted signal may be. Moreover, the timing of foveal feedback effects during fixation varied considerably across previously published investigations (from 117 ms in Weldon et al. (2016) to 2020) to 550 ms in Fan et al. (2016)). While this variation has been ascribed to a flexible adjustment of the feedback mechanism to current task demands (Fan et al., 2016), low-level stimulus features may play an equally important role, at least in our investigations. In particular, neuronal response latencies decrease systematically as the contrast of visual input increases. While this phenomenon is reliably observed at varying stages of the visual processing hierarchy—such as the lateral geniculate nucleus (Lee, Elepfandt, & Virsu, 1981b), primary visual cortex (e.g., Albrecht, 1995; Carandini & Heeger, 1994; Carandini, Heeger, & Movshon, 1997; Carandini, Heeger, & Senn, 2002), and anterior superior temporal sulcus (STSa; Oram, Xiao, Dritschel, & Payne, 2002; van Rossum, van der Meer, Xiao, & Oram, 2008)—influences of contrast on neuronal response latency are particularly pronounced in higher-order visual areas: A doubling of stimulus contrast has been shown to decrease the latency of V1 neurons by 8 ms, compared to a reduction of 33 ms in area STSa (Oram et al., 2002; van Rossum et al., 2008). Assuming that the peripheral target is processed in a bottom-up fashion until it reaches higher-order object processing areas, the time point at which peripheral signals are available for feedback should be dictated by the temporal dynamics of visual feedforward processing.

In this Research Advance, we describe the influence of saccade target opacity (throughout the paper, we use the term ‘opacity’ when we refer to the experimental manipulation (that is, a variation of the transparency, i.e., 1-opacity of the target patch against the background noise) and the term ‘conspicuity’ when we discuss our findings conceptually) on pre-saccadic foveal congruency effects. For this purpose, we varied target opacity in four steps (equally spaced on a log scale: 25.0%, 38.3%, 58.7%, 90.0%) while leaving the remaining task parameters unchanged (Figure 1). An increase in target opacity from 25% to 90% corresponded to an increase in Michelson contrast (Michelson, 1927) from .53 to .78 and an increase in the SNR of the filtered orientation from 1.40 to 2.58 (see **Methods**). Note that the foveal probe was presented at the same opacity (individually adjusted for every session and observer) across all target opacity conditions. The difference between congruent and incongruent HRs for foveal probes, i.e., enhancement, increased monotonically with target opacity. This finding endorses the assumption that predictive foveal processing can support perceptual continuity in natural visual environments, in which saccades are routinely directed towards the currently most conspicuous object in the scene (‘t Hart, Schmidt, Roth, & Einhäuser, 2013). Moreover, especially when systematic variations of saccade latency were taken into account, enhancement (note that throughout the manuscript, the term ‘enhancement’ is used to refer to the difference between congruent and incongruent Hit Rates [$HR_{\text{cong}} - HR_{\text{incong}}$]) required less and less time to develop as the opacity of the saccade target, and thus its contrast, increased. These findings not only provide new mechanistic insights but refine our method and are relevant for researchers planning to use or adapt our paradigm to study related questions.

Results

Hit and False Alarm rates as a function of target opacity

Both congruent and incongruent HRs decreased with increasing target opacity (Figure 2A, left panel), arguably because attentional resources were increasingly drawn to the target the more salient it became. This decrease in foveal HRs, however, was less pronounced for probes with target-congruent orientation [slope of the linear regression line: -0.46% (congruent) vs -1.73% (incongruent) per opacity step, $p < .001$]. In all four opacity conditions, observers detected foveal probes more readily if they exhibited the same orientation as the saccade target, resulting in significantly higher congruent as compared to incongruent HRs ($HR_{\text{cong-incong}} = 3.3\%, 3.4\%, 5.4\% \& 6.8\%$, $ps < .021$; Figure 2A, middle panel). To fully characterize the impact of target opacity on enhancement in HRs, we performed a linear mixed-effects model in which we described the

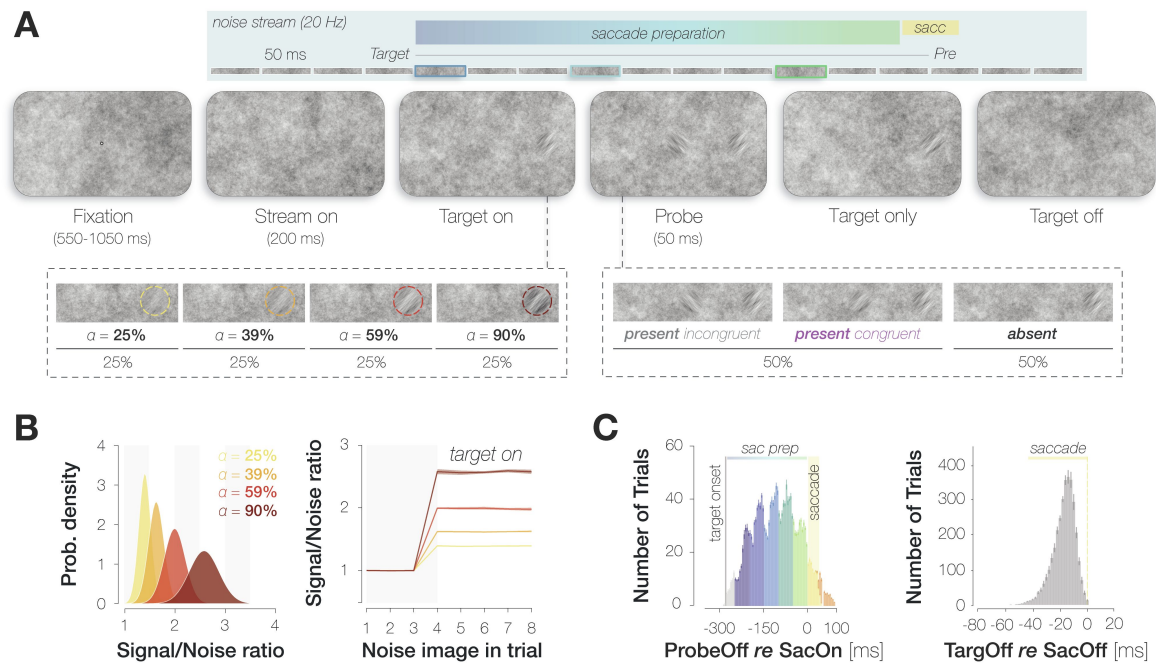


Figure 1.

Summary of the paradigm.

A. Example trial procedure: The saccade target and foveal probe were embedded in full-screen noise images flickering at a frequency of 20 Hz (image duration of 50 ms). After 200 ms, the saccade target (an orientation-filtered patch; filtered to either -45° or $+45^\circ$; 3 dva in diameter) appeared 10 dva to the left or the right of the screen center, cueing the eye movement. On 50% of trials, a probe (a second orientation-filtered patch; filtered to either -45° or $+45^\circ$) appeared in the screen center at an early (top panel; highlighted element with dark blue outline), medium (light blue outline) or late (green outline) stage of saccade preparation. The foveal probe was presented for 50 ms and could be oriented either congruently or incongruently to the target. In contrast to our previous investigation, the saccade target was presented at one of four opacities (from 25% to 90%; in different blocks). **B.** An increase in target opacity translates to an increase in signal-to-noise ratio (SNR; left panel). Within a single trial, the increase in SNR at the target location manifests from the fourth noise image on (i.e., after the target appears; right panel; error bands correspond to the standard deviation across all images). **C.** Probe and target timing. Probe timing (left panel): histogram of time intervals between probe offset and saccade onset. Bar heights and error bars indicate the mean and standard error of the mean (SEM; $n=9$) across observers, respectively. On included trials, the probe appeared after target onset and therefore during saccade preparation ('sac prep'). Trials in which the probe disappeared more than 250 ms before saccade onset (light grey), during the saccade (yellow) or after saccade offset (orange) were excluded. The yellow background rectangle illustrates the median saccade duration. Target timing (right panel): histogram of time intervals between target offset and saccade offset. Bar heights and error bars indicate the mean and standard error of the mean (SEM; $n=9$) across observers, respectively. Unlike in the previous study, we removed the target upon saccade initiation on all trials.

variance in enhancement ($HR_{\text{cong}} - HR_{\text{incong}}$) across all time points with a fixed effect of target opacity and a random intercept for observer (see **Methods**). Note that this model outperformed the simplest model including a fixed effect of target opacity only ($\Delta BIC = 11.6$) and a more complex one involving a random intercept and random slope for observer ($\Delta BIC = 4.4$). The fixed effect of target opacity reached significance, $t(34) = 2.341$, $p = 0.025$. More specifically, as the opacity of the saccade target increased, the difference between HRs for target-congruent and target-incongruent foveal probes increased as well, resulting in a significantly positive slope of the fitted linear regression line (slope = 1.3% per opacity step, $p < .001$; **Figure 2A**, right panel).

On probe-absent trials, observers reported perceiving the target orientation more often than the non-target orientation for the 25%, 59% and 90% opacity condition, resulting in significantly higher congruent as compared to incongruent False Alarm Rates (FARs; $FAR_{\text{cong-incong}} = 6.5\%$, 6.2% & 4.6% , $ps < .02$; **Figure 2B**). We performed a linear mixed-effects model to describe the difference between congruent and incongruent FARs with a fixed effect of target opacity and a random intercept for observer. Unlike for HRs, the fixed effect of target opacity did not affect the differences in FARs, $t(34) = -0.44$, $p = .661$. The slope of the fitted linear regression line was statistically indistinguishable from zero (-0.2% per opacity step, $p = .713$). Inspecting congruent and incongruent FARs instead of their difference revealed that, just like HRs, they both decreased with increasing target opacity. This decrease, however, was comparable for congruent and incongruent FARs (slopes: -1.9% vs -1.6% per opacity step, $p = .715$).

In our paradigm, the entire screen was covered with pink noise. In consequence, the display region the probe could appear in was never void of signal but contained incidental orientation information even on probe-absent trials. Trials in which observers reported perceiving the probe when no stimulus was presented may therefore contain a mixture of two response types: (i) ‘true’ FAs, that is, unsystematic or biased reports and (ii) ‘true Hits’, that is, systematic reactions to incidental orientation information in the foveal noise region. As target opacity (and, presumably, the strength of the feedback signal) increased, the proportion of ‘true Hits’ within the subset of congruent FAs may have increased while the proportion of the ‘true FAs’ decreased. The relative contribution of both response types may thus have varied considerably without affecting the mean congruent FAR. Indeed, two observations suggest that the contribution of unsystematic responses to FAs in general, and congruent FAs in particular, decreased as target opacity increased.

First, we described the visual properties of the foveal noise on FA trials by convolving a set of Gabor filters with different spatial frequencies (SFs) and orientations with the pixel content in a 3 dva region around the pre-saccadic center of gaze. We then subtracted filter energies on incongruent FA trials from filter energies on congruent FA trials (see **Methods**). If FAs relied partially on signal, this difference image should be characterized by high filter energies around the target orientation and low filter energies around the non-target orientation. Interestingly, the difference in filter energies around the target and non-target orientation was significantly more pronounced for the higher as compared to the lower two target opacities ($p = .008$; **Figure 2C**). This observation suggests that the contribution of ‘true Hits’ to the pool of foveal FAs increased with target opacity, allowing systematic orientation effects to manifest in average noise properties.

Second, we related the dependence of HRs on target opacity to the dependence of FARs on target opacity across observers. Specifically, we correlated the z-standardized slope of the difference in HRs (**Figure 2A**, right panel) with the z-standardized slope of the difference in FARs (**Figure 2B**, right panel). This analysis yielded a moderate yet non-significant negative correlation between HR and FAR slopes ($r = -.41$, $p = .278$). In other words, observers who showed more enhancement in HRs as target opacity increased tended to show a smaller difference between congruent and incongruent FARs with increasing opacity. This correlation is based on a small number of observers ($n = 9$) and relates two difference measures which can be considered noisy

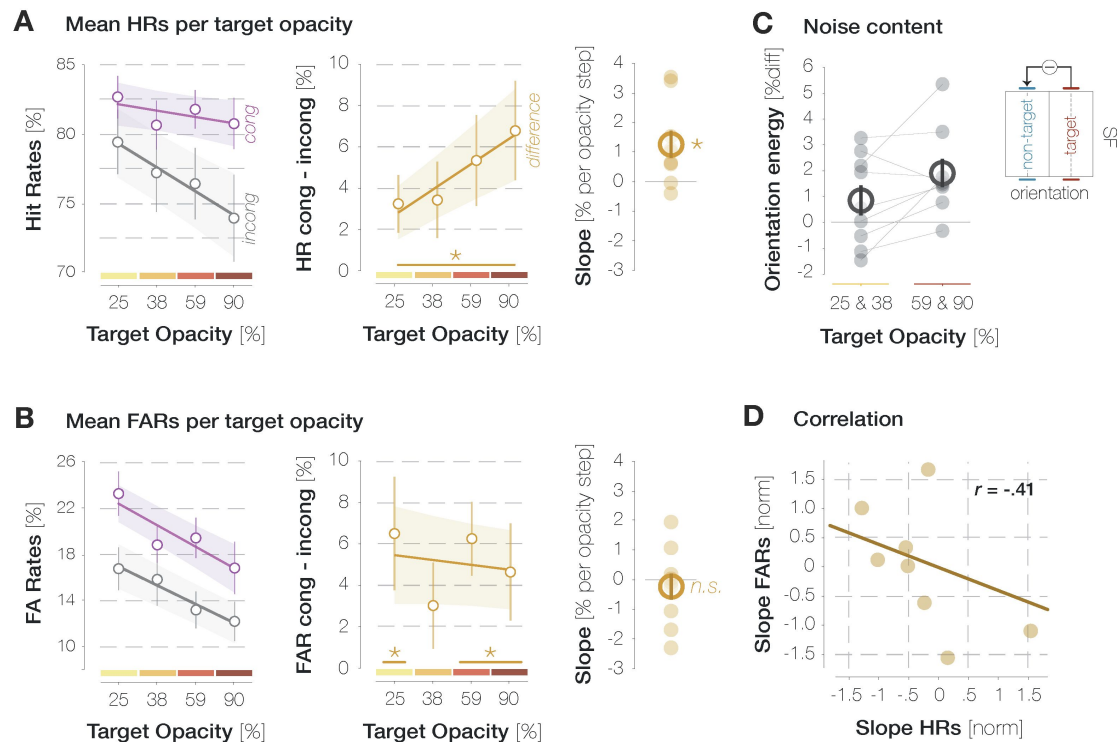


Figure 2.

Influence of target opacity on Hits and FAs across all time points.

A. Influence of target opacity on congruent and incongruent HRs (purple and gray data points in the left panel) as well as their difference (orange data points, middle panel). Lines and error bands correspond to the fitted linear regression lines ± 2 standard errors of the mean (SEM). The slopes of the fitted regression line per observer (small circles) and their mean and SEM (big circle and error bar) are plotted in the right panel. Asterisks denote statistically significant comparisons ($p < .05$; determined via bootstrapping; $n = 9$ observers). **B.** Influence of target opacity on congruent and incongruent FARs (left panel) as well as their difference (middle and right panel). All conventions are as in A. **C.** Mean difference in filter energies around the target and non-target orientation for the lower two target opacities (left column) and the higher two target opacities (right panel). Lines connect the values of individual observers (small circles) in both conditions. Large circles and error bars denote the mean and SEM, respectively. **D.** Pearson correlation between the normalized slope in HRs (A, right panel) and the normalized slope in FARs (B, right panel). Circles indicate individual observers.

estimates to begin with. It should thus be validated in future investigations involving a larger sample. Nonetheless, it suggests that higher target opacity levels increased observers' sensitivity for congruent foveal orientation signals.

After establishing the impact of target opacity on HRs and FARs across all time points, we aimed to characterize the influence of target opacity on the time course of foveal enhancement. Yet, we expected target opacity to influence saccade latencies in a systematic fashion, potentially complicating a direct comparison of time courses between opacity conditions. Three interdependent factors are likely to influence the time course of congruency effects in HRs: (1) the duration for which the target had been visible when the foveal probe appeared (target-locked time course), (2) the stage of saccade preparation at which the probe was presented (saccade-locked time course) and (3) the latency of a given eye movement which moderates the relation between (1) and (2). In consequence, a probe stimulus presented, for instance, 150 ms after the target will appear in later and later stages of saccade preparation as saccade latency decreases.

Saccade parameters

Median saccade latencies decreased as the opacity and, therefore, the contrast of the saccade target increased (median *mdn* = 288.8, 279.7, 276.4, & 268.0 ms; **Figure 3A**). The slope of the linear regression line fitted to median latencies on an individual-observer level was significantly smaller than zero (mean *m* = -6.57 ms/opacity step, *p* < .001). Target opacity did not affect the precision of saccadic timing: the slope of the regression line fitted to the standard deviation of latencies on an individual-observer level was not significantly smaller than zero (*m* = -0.41 ms/opacity step, *p* = .169). To fully characterize the influence of target opacity on saccade metrics, we computed bivariate Gaussian kernel densities of observers' saccade landing coordinates per opacity (**Supplemental Figure S1B**). The center of mass fell inside the target region in all conditions, demonstrating that observers were able to execute accurate saccades even at the lowest opacity. Indeed, saccade amplitudes were uninfluenced by target opacity (*mdn* = 10.20, 10.31, 10.25 & 10.33 dva): The slope of the regression line fitted to median amplitudes on an individual-observer level was statistically indistinguishable from zero (*m* = 0.03 dva/opacity step, *p* = .153, **Supplemental Figure S1D**, panel 2). Likewise, saccadic error defined as the Euclidean distance between saccade landing coordinates and the center of the target stimulus did not vary with target opacity (*mdn* = 1.35, 1.35, 1.38 & 1.37 dva, *p* = .171; **Supplemental Figure S1D**, panel 3). Saccadic peak velocities, on the other hand, decreased significantly with opacity (*mdn* = 416.7, 413.70, 410.17, 407.25 dva/s, *p* < .001; **Supplemental Figure S1D**, panel 4). However, due to the small absolute scale of these variations, the main sequence defined as the relation between saccade amplitudes and peak velocities remained largely unaltered (**Supplemental Figure S1C**).

The influence of target opacity on the time course of enhancement

When probe onset was temporally aligned to the **onset of the peripheral target stimulus** (**Figure 3B**; **first row**), congruent and incongruent HRs did not differ significantly in any time bin for the lowest two opacities (all *ps* between .085 and .686; obtained with bootstrapping, see **Methods**). For targets with 59% opacity, congruent HRs significantly exceeded incongruent ones if the probe appeared 100-150 or 200-300 ms after the target (*p* = .003 & *p* < .001). For the highest target opacity, we observed a continuous enhancement window ranging from 100 to 200 ms after target onset (*ps* < .003). We subsequently inspected the saccade-locked time course by temporally aligning the offset of the probe stimulus to the **onset of the eye movement**. The saccade-locked development of foveal enhancement roughly mirrors target-locked time courses (**Figure 3B**; **second row**). Yet, due to the systematic shortening of saccade latencies with target opacity (see **Figure 3A**), the same saccade-locked time bin contained trials with systematically different target-probe intervals for different target opacities. In the last pre-saccadic bin, for instance, the probe had appeared 211 ms after the target for the lowest opacity and 178 ms after the target for the highest opacity. To control for the influence of target-probe intervals on saccade-locked time courses, we determined

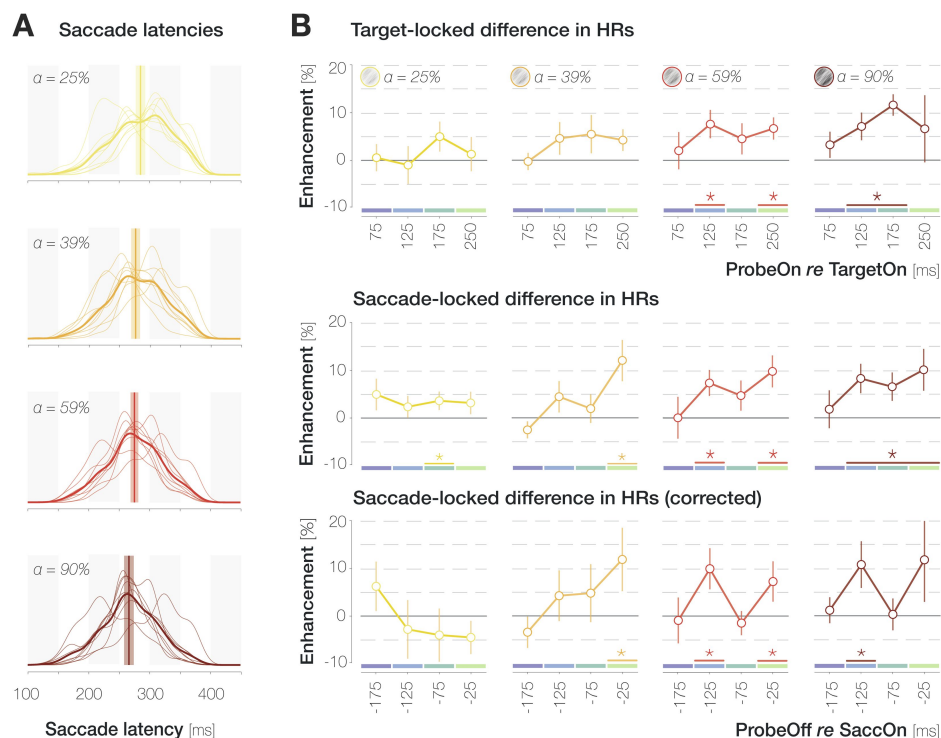


Figure 3.

Time course of enhancement in HRs for different target opacity levels.

A. Probability density distributions of saccade latencies for increasing target opacity. Distributions with thin and thick lines represent individual-observer and mean probability densities, respectively. Vertical lines and shaded regions represent median latencies and SEMs, respectively. **B.** Target- (first row) and saccade- (second and third row) locked time course of enhancement. To obtain the corrected saccade-locked time course, the proportion of different target-locked bins in each saccade-locked bin was equalized across opacities to account for the systematic decrease in latencies with increasing opacity (see A). X-axis values indicate the center of 50 ms bins. Note that for saccade-locked time courses, the last bin contains probe onset times between 200 and 300 ms to allow for sufficient trial numbers. Across panels, error bars indicate SEMs. Asterisks denote significant differences between congruent and incongruent HRs ($p \leq .05$; determined via bootstrapping with 10,000 repetitions; $n = 9$ observers).

the proportion of each target-probe offset in every pre-saccadic time bin. We then equalized these proportions across target opacities. To achieve this, we expressed the proportions in the 59% opacity condition (i.e., approximately the opacity used in our previous investigations) as multiples of the proportions in every remaining opacity condition. We subsequently weighted the saccade-locked enhancement values by these relative proportions (see Rolfs & Carrasco, 2012 for a similar approach). As a result, enhancement in each saccade-locked bin in **Figure 3C** is reconstructed to contain similar probe-target offsets across opacities.

We did not observe a significant difference between congruent and incongruent HRs in any pre-saccadic time bin for the lowest target opacity (all p s between .095 and .905). At an opacity of 39%, significant enhancement manifested during the 50 ms immediately preceding saccade onset, $p = .006$. As target opacity increased further, enhancement was additionally observable in an earlier saccade preparation bin from 150 to 100 ms before eye movement onset, $p = .003$. At the highest target opacity, congruent HRs exceeded incongruent ones in the 150-100 ms bin exclusively, $p = .005$. In short, the corrected saccade-locked time courses appear to suggest that foveal enhancement is observable in earlier stages of saccade preparation as target opacity increases.

Especially at higher target opacities, the temporal development of foveal enhancement appears to exhibit an oscillatory pattern. To inspect this incidental observation in a more temporally resolved fashion, we determined mean enhancement values in a boxcar window of 50 ms duration sliding along all saccade-locked probe offset time points (step size = 10 ms; x-axis values in **Figure 4** indicate the latest time point in a certain window). We then fitted 6th order polynomials (with no constraints on parameters) to the resulting time courses and determined significant time points using bootstrapping (see Methods). The average foveal enhancement across target opacities reached significance starting 115 ms before saccade onset (gray curve in **Figure 4**; all p s < .046). For every individual target opacity condition, we observed significant enhancement immediately before saccade onset, although only very briefly for the lowest opacity (−2–0 ms for 25%; −39–0 ms for 39%, −106–0 ms for 59% & −13–0 ms for 90%; all p s < .050; yellow to dark red curves in **Figure 4**). Especially for the higher two target opacities, we observed a local maximum preceding eye movement onset by approximately 80 ms. Interestingly, assuming a peak in enhancement in approximately 80 ms intervals (i.e., at x-axis values of −80 and 0 ms in **Figure 4**) would correspond to an oscillation frequency of 12.5 Hz. In contrast to rapid feedforward processing, feedback signaling is associated with neural oscillations in the alpha and beta range (i.e., between 7 and 30 Hz; Bastos et al., 2015; Jensen, Bonnefond, Marshall, & Tiesinga, 2015; van Kerkoerle et al., 2014).

The influence of saccade latency on foveal congruency effects

On an individual-trial level, the latency of the eye movement may impact foveal detection judgments beyond influencing probe timing. Since the saccade target was visible throughout saccade preparation, peripheral orientation information could accumulate for a longer period of time for long-latency saccades and may have consequently exerted a larger influence on foveal detection. On the other hand, trials with short saccade latencies likely reflect instances in which the saccade target was localized with minimal spatial uncertainty and without erroneous attentional allocation to the opposite hemifield. If short saccade latencies rely on effective target selection and oculomotor planning (see also Jonikaitis & Theeuwes, 2013; Jonikaitis, Klapetek, & Deubel, 2017; Yan, Zhaoping & Li, 2018), congruency effects may emerge most robustly on these trials. To inspect the influence of saccade latency on foveal HRs and FARs, we determined each observer's median saccade latency per target opacity. We then separated all trials into short-latency ($md = 247 \pm 18$ ms) and long-latency ($md = 308 \pm 18$ ms) subsets, depending on whether the latency of the executed saccade on a given trial was shorter or longer than an observer's median latency for the respective opacity level.

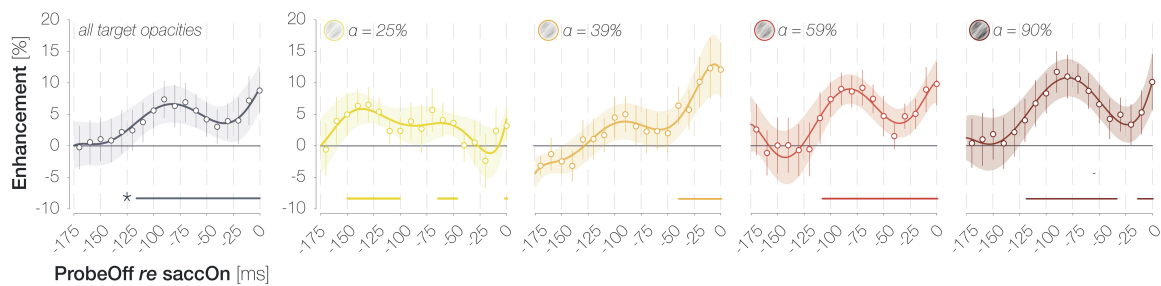


Figure 4.

Continuous time course of enhancement in HRs (HRcong–HRincong) across target opacities (gray; plot 1) and separately for different target opacity levels (plots 2–5; yellow to dark red).

X-axis values indicate the latest time point in a sliding boxcar window of 50 ms duration. Data points and error bars correspond to the mean ± 1 SEM across observers. Lines and error bands correspond to mean 6th order polynomial fits ± 1 SEM. Horizontal lines above the x-axes denote significant enhancement ($p \leq .05$; determined via bootstrapping with 10,000 repetitions; $n = 9$ observers).

All previously described effects were amplified in the subset of short-latency saccades (mean $n = 1,623$ trials per observer): Across all time points, the difference between congruent and incongruent HRs reached significance for target opacities $\geq 39\%$ and increased with opacity (slope = 2.85% per opacity step, $p < .001$; **Figure 5A**). The difference in FARs, in turn, did not vary significantly with opacity (slope = 0.64% per opacity step, $p = .883$). Again, we observed a moderately negative yet non-significant correlation of $r = -.49$ ($p = .181$) between the normalized differences in HRs and FARs.

Long-latency saccades, in contrast, showed a markedly different pattern (mean $n = 1,218$ trials per observer): Enhancement in HRs was overall less pronounced and even decreased with increasing target SNR (slope = -1.23% per SNR step, $p = .044$; **Figure 5B**). The differences in HRs and FARs were entirely uncorrelated ($r = .09$, $p = .823$). These findings suggest that short-latency saccades, which were enabled by effective attentional target selection and oculomotor planning, had driven the enhancement effects reported for the entire trial pool (see **Figure 2**). The small but significant decrease in enhancement with increasing target opacity in the long-latency subset could suggest that long-latency saccades to high-contrast targets in particular rely on attentional lapses which affected pre-saccadic perceptual processes as well as oculomotor behavior.

Discussion

We investigated whether pre-saccadic foveal congruency effects demonstrated in a previous investigation (Kroell & Rolfs, 2022) are influenced by the conspicuity of the eye movement target stimulus. For this purpose, we varied the opacity of the orientation-filtered saccade target against the unfiltered 1/f background noise. Along with the SNR of target-like orientation information, this manipulation systematically altered the luminance contrast of the target patch (see **Methods**). Four observations can be highlighted.

First, the manipulation of target opacity influenced saccade metrics: Saccade latencies as well as saccadic peak velocities decreased with increasing opacity. The decrease in latencies was to be expected and likely reflects a facilitation of movement planning towards high-contrast target stimuli (Ludwig, Gilchrist, & McSorley, 2004). The small yet significant decrease in peak velocities which cannot be explained by a concomitant decrease in saccade amplitudes was unexpected and, to the best of our knowledge, has not been observed previously. Hypothetically, higher peak velocities may serve to compensate for longer saccade latencies by slightly decreasing the overall time interval between target appearance and saccade landing.

Second, HRs for target-congruent and incongruent foveal probes decreased as target opacity increased, likely because attention was increasingly drawn to the target the more salient it became. Crucially, this decrease was less pronounced for congruent than for incongruent HRs, resulting in a continuous increase of enhancement with target opacity. From an experimenter's perspective, presenting the target at a high opacity (59% and above in our design) seems beneficial since congruency effects, especially when time-resolved, are more robustly detectable. On a more conceptual level, pre-saccadic attention seems to select the target orientation in an automatic fashion, even when the eye movement could easily be programmed based on local contrast variations alone. The influence of peripheral orientation information on foveal detection judgments is directly proportional to the strength of the orientation signal at the target location. As during spontaneous eye movement behavior, saccades are routinely directed towards the currently most conspicuous object in the scene (t Hart et al., 2013), this finding underscores the feasibility of predictive foveal processing in natural visual environments. Of note, the relation between target opacity and foveal enhancement was exclusively driven by saccades with latencies below an observer's median latency in each opacity condition. While the individual-trial relationship between saccade latency and the effect size of pre- and transsaccadic mechanisms is seldomly explored (but see Jonikaitis & Deubel, 2011; Jonikaitis & Theeuwes, 2013; Jonikaitis et

al., 2017 [↗](#); Yan et al., 2018 [↗](#)), we suggest that short saccade latencies reflect effective target selection and oculomotor planning which in turn benefits pre-saccadic perceptual processes such as foveal prediction.

Third, target opacity influenced the time course of enhancement in HRs. When systematic variations of saccade latency were taken into account (**Figure 3B** [↗](#); third row), congruent and incongruent HRs differed earlier for higher target opacities. These findings suggest that the feedforward processing of the peripheral saccade target may have been accelerated when it was presented at high contrast (Albrecht et al., 2002 [↗](#)). Alternatively, or in addition, it is conceivable that weaker feedforward signals require a longer accumulation interval before the feedback process can be initiated. Moreover, foveal congruency effects appear to exhibit an oscillatory pattern, with peaks in a medium saccade preparation stage (~80 ms before the eye movement) and immediately before saccade onset. We have noticed this pattern in several investigations with substantially different visual stimuli and behavioral readouts. For instance, using a full-screen dot motion paradigm, we observed a pre-saccadic, small-gain ocular following response to coherent motion in the saccade target region (Kroell, Rolfs, & Mitchell, 2023, conference abstract; Kroell, 2023, dissertation). Predictive ocular following first reached significance ~125 ms before the eye movement, then decreased and subsequently ramped up again ~25 ms before saccade onset. Several explanatory mechanisms appear conceivable. Unlike rapid feedforward processing, feedback propagation has been shown to follow an oscillatory rhythm in the alpha and beta range, that is, between 7 and 30 Hz (Bastos et al., 2015 [↗](#); Jensen, Bonnefond, Marshall, & Tiesinga, 2015 [↗](#); van Kerkoerle, et al., 2014 [↗](#)). In our case, it is possible that the object-processing areas that send feedback to retinotopic visual cortex do so at a temporal frequency of ~12.5 Hz. At higher stimulus contrasts, feedforward signals may be fed back instantaneously and without the need for signal accumulation in feedback-generating areas. The resulting perceptual time courses may reflect innate temporal feedback properties most veridically. Alternatively, the initial enhancement peak may be related to the sudden onset of the saccade target stimulus and not to movement preparation itself. In this case, the initial peak should become particularly apparent if enhancement is aligned to the onset of the target stimulus. Yet, **Figure 3** [↗](#) and **Figure 4** [↗](#) suggest more prominent oscillations in saccade-locked time courses. In accordance with this, perceptual and attentional processes have been shown to exhibit oscillatory modulations that are phase-locked to action onset (e.g., Tomassini, Spinelli, Jacono, Sandini, & Morrone, 2015; Hogendoorn, 2016; Wutz, Muschler, van Koningsbruggen, Weisz, & Melcher, 2016; Benedetto & Morrone, 2017; Tomassini, Ambrogioni, Medendorp, & Maris, 2017; Benedetto, Morrone & Tomassini, 2019). Whether the oscillatory pattern of foveal enhancement, as well as its increased prominence at higher target contrasts, relies on innate temporal properties of feedback signaling, signal accumulation, saccade-locked oscillatory modulations of feedforward processing or attention, or a combination of these factors, one conclusion remains: task-induced cognitive influences suggested to underlie the considerable variability in temporal characteristics of foveal feedback during passive fixation (e.g., Fan et al., 2016 [↗](#); Weldon et al., 2016 [↗](#); 2020 [↗](#)) are not the only possible explanation. Low-level target properties such as its luminance contrast modulate the resulting time course and should be equally considered, at least in our paradigm.

Fourth, unlike the difference between congruent and incongruent HRs, the difference in FARs did not increase with target opacity. Nonetheless, we suggest that the contribution of different response types to the pool of FAs varied across opacities: As opacity (and, therefore, the difference in HRs) increased, incidental visual properties in the foveal background noise more reliably reflected the subsequently reported orientation. In other words, FAs were increasingly triggered by signal, that is, orientation information presented on screen, as target opacity increased. Moreover, observers who showed more enhancement in HRs as target opacity increased tended to show a smaller difference between congruent and incongruent FARs with increasing opacity. Combined with a range of arguments provided in our previous publication (Kroell & Rolfs, 2022 [↗](#)), this observation suggests that foveal congruency effects reflect a variation of pre-saccadic

foveal sensitivity rather than a shift in a post-perceptual decision criterion. As noted in the **Results** section, however, this finding is based on a small number of observations and should be confirmed in larger samples.

Foveal prediction in natural visual environments

As noted above, human observers typically move their eyes towards the most conspicuous objects in their environment. Foveal prediction seems to benefit from this strategy as the strength of the predicted signal increases with the conspicuity of the eye movement target. Nonetheless, natural visual environments as well as natural viewing behavior pose several challenges for the foveal prediction mechanism (see Kroell & Rolfs, 2022 [↗](#), for an initial discussion).

First, naturalistic saccade target stimuli will likely exhibit complex shapes and, more often than not, will include feature conjunctions rather than isolated features. Previous findings suggest that the foveal feedback mechanism is capable of operating at this level of complexity: High-level peripheral information such as the category of novel, rendered objects has been successfully decoded from activation in foveal retinotopic cortex (e.g., Williams et al., 2008 [↗](#)). If, indeed, temporal object-specific areas such as area TE send feedback, the foveal prediction mechanism may even be specialized for the transfer of complex visual properties.

Second, foveal input will often be of high contrast in natural visual environments. If fed-back predictive signals can influence foveal perception in the presence of high-contrast feedforward input remains to be established. In our main investigation (Kroell & Rolfs, 2022 [↗](#); **Figure 2B** [↗](#)) as well as in previous studies (Hanning & Deubel, 2022b [↗](#)), pre-saccadic foveal detection performances decreased markedly in the course of saccade preparation, presumably because visuospatial attention gradually shifted towards the saccade target and away from the foveal location. This pre-saccadic decrease in foveal sensitivity may boost the relative weight of fed-back signals by attenuating the conspicuity of high-contrast feedforward input. In other words, the strength of feedforward input to the fovea is reduced gradually across saccade preparation. At the same time, the strength of the fed-back predictive signal should profit from the high contrast of naturalistic saccade targets.

Third, while foveal and peripheral information was congruent on 50% of all ‘probe present’ trials in our investigation, peripheral and foveal features will often be weakly correlated (see Samonds, Geisler, & Priebe, 2018) or even uncorrelated in natural environments. Again, the pre-saccadic attenuation of foveal feedforward processing may allow fed-back peripheral signals to influence perception even if they are uncorrelated with foveal information. Moreover, in piloting variations of our paradigm, we observed that the subjective impression of perceiving the saccade target at the pre-saccadic foveal location is most pronounced if the foveal noise region is replaced with a black Gaussian blob at certain time points before saccade onset (observation by author LMK; unpublished). In consequence, fed-back signals do not seem to require correlated feedforward input to influence perception. Quantitative evidence, however, remains to be established.

Lastly, pre-saccadic foveal input is likely less relevant during natural viewing behavior than it is in our task. It is possible that this task-induced prioritization of the foveal location facilitated the emergence of congruency effects. In a previous experiment (Kroell & Rolfs, 2022 [↗](#); **Figure 1D** [↗](#)), however, the perceptual probe could appear anywhere on a horizontal axis of 9 dva length around the screen center. Despite this spatial unpredictability, however, congruency effects peaked at the pre-saccadic foveal location, even after peripheral baseline performances had been raised to a foveal level through an adaptive increase in probe opacity. Ultimately, an influence of task demands on visual processing can only be fully excluded through techniques that provide a direct readout of perceptual contents without requiring keyboard responses. In psychophysical investigations, a prediction of saccade target motion may be read out from observers’ eye velocities (Kroell, Mitchell, & Rolfs, 2023; Kwon, Rolfs, & Mitchell, 2019). In

electroencephalographic (EEG) and neurophysiological studies, foveal predictions should manifest in early visually evoked potentials (e.g., Creel, 2019) and increased firing rates of feature-selective foveal neurons in early visual areas, respectively.

Neural implementation of foveal prediction

Based on the body of our findings as well as previous literature, we suggested a parsimonious feedback mechanism to underly the observed effects: the preparation of a saccadic eye movement, and the concomitant shift of pre-saccadic attention (e.g., Kowler, Anderson, Doshier, & Blaser, 1995 [↗](#); Deubel & Schneider, 1996 [↗](#)), selects the peripheral target stimulus among competing information. Higher-order visual areas feed selected feature input back to early retinotopic areas—specifically, to neurons with foveal receptive fields. Fed-back feature information combines with congruent, foveal feedforward input, resulting in the enhancement effects we observe. Especially in the context of active vision, this feedback mechanism is appealing as it resolves a combinatorial issue associated with feature-specific information transfer before saccades. Consider a simplified case in which, right before a saccadic eye movement, the activation of a feature-selective neuron that encodes a certain retinal location is transferred to a neuron within the same brain area that will encode said retinal location after saccade landing. For this mechanism to function for any possible saccade direction and amplitude, most neurons would need to be connected to most other neurons (or, in a simplified version, to neurons with foveal receptive fields) in a given brain area. Assuming an information transmission via feedback rather than horizontal connections significantly reduces this dimensionality: Higher-order visual areas that encode object properties (largely) detached from retinotopic or spatiotopic reference frames selectively transfer feature information to neurons with foveal receptive fields, irrespective of the vector of the upcoming saccade. This parsimonious mechanism would have shortcomings. In particular, foveal feedback should become less effective during saccade sequences where several peripheral targets are simultaneously attended. Feature information at both attended target locations may be fed back in temporal succession or weighted and erroneously combined into a single fed-back signal. In most cases, however, foveal feedback may reasonably achieve what established transsaccadic mechanisms struggle to explain: An anticipation of the features of a single saccade target—which typically constitutes the currently most relevant object in the visual field—in foveal vision.

While direct feedback connections from higher-order to early visual areas would constitute the most straightforward implementation, it is conceivable that feedback signals are relayed through and modulated by subcortical areas. In particular, the thalamic pulvinar has been identified as a connection hub for visual processing that receives copies of feedforward and feedback connections from different visual areas and may even combine information across visual space (Cortes, Ladret, Abbas-Farishta, & Casanova, 2024 [↗](#)). In the case of foveal prediction, thalamic neurons may receive fed-back signals from higher-order areas and enhance those signals before passing them on to cortical neurons with foveal receptive fields. Perhaps, a modification of foveal activation within the thalamic pulvinar itself is sufficient to influence perception. To the best of our understanding, however, the fed-back signal must originate in non-retinotopic, higher-order object processing areas to reduce the number of necessary neuronal connections.

Conclusion

To sum up, this investigation both provides further mechanistic insights into the pre-saccadic foveal prediction mechanism and constrains the parameter space for researchers planning to adapt our paradigm in future studies. Conceptually, active peripheral feature sampling is not necessary for foveal prediction to emerge. Instead, foveal congruency effects develop even when (or especially when) salient local contrast variations at the saccade target location can be used to direct the eye movement. In consequence, presenting the target at a high opacity appears purely

beneficial. At high target opacities, foveal congruency effects in HRs are more pronounced. Moreover, observers respond more systematically to target-like orientations in foveally presented noise, facilitating reverse-correlation based analyses.

Methods

Sample

Nine human observers (six females, eight right-handed, six right-eye dominant, no authors) aged 22 to 30 years ($mdn = 26.0$) participated in the experiment. Normal ($n = 3$) or corrected-to normal ($n = 6$) visual acuity was ensured at the beginning of the first session using a Snellen chart (Hetherington, 1954 [↗](#)) embedded in a Polatest vision testing instrument (Zeiss, Oberkochen, Germany). Observers yielding scores of 20/25 or 20/20 were invited to proceed with the experiment. Ocular dominance was assessed using the Miles test (Miles, 1930 [↗](#)). Since data collection was performed during the COVID-19 pandemic, our sample was composed of lab members. Nonetheless, all observers were naïve as to the purpose of the study. Participants gave written informed consent before the experiments and were compensated with either accreditation of work hours or a payment of 8.50€/hour plus a bonus of 1€/session. The study complied with the Declaration of Helsinki in its latest version and was approved by the Ethics Committee of the Department of Psychology at Humboldt-Universität zu Berlin. The research question, experimental paradigm and data analyses were preregistered on the Open Science Framework (*link will be made available upon acceptance*). Raw and pre-processed data as well as the experimental code are available in the same repository.

Experimental setup and design

The external setup, session structure and trial time course were identical to our previous experiments (Kroell & Rolfs, 2022 [↗](#)) with the following exceptions: First and most importantly, the saccade target was overlayed on the background noise with one of four logarithmically spaced opacities ($\alpha = 25\%$, 39% , 59% and 90% compared to 60% in Kroell & Rolfs, 2022 [↗](#)). Targets with different opacities were presented in separate experimental blocks the order of which was randomized. Before every block, we showed a preview of the target stimulus at the respective opacity (sequentially at both possible target locations in the left and right screen half, randomly oriented either 45° to the left or right). Second, during each trial in a block, we removed the target upon saccade initiation, i.e., once gaze had crossed a boundary of 2.0 dva diameter around the fixation dot. Third, we increased the number of potential background noise images from 17 to 27 images per session and observer. All 27 images were generated at the beginning of a given session. Before each trial, we randomly drew 16 images from this pool (without replacement) and presented them in random order. Since 9 observers each completed 7 sessions, $27 \times 9 \times 7 = 1701$ different noise images were presented in the course of the experiment. Lastly, observers completed four staircase blocks before the main experiment in every session (one block per target opacity). Individual staircase blocks were short (32 trials each) and served to familiarize observers with all target opacities as well as to obtain an estimate of observers' typical saccade latency for each opacity. Saccade latency estimates were used to define three possible probe onset times for each target opacity in the main experiment. Importantly, however, the same staircase ran throughout all four blocks and returned one optimal probe opacity estimate that was used throughout the main experiment, irrespective of target opacity.

Increasing target opacity increased the luminance contrast and the SNR of the filtered orientation within the saccade target region. During the experiment, we manipulated target opacity by varying the 'globalAlpha' input to the Psychtoolbox function 'DrawTexture' (see <http://psychtoolbox.org/docs/Screen-DrawTexture> [↗](#)) and enabled alpha blending with the inputs 'GL_SRC_ALPHA' (source) and 'GL_ONE_MINUS_SRC_ALPHA' (destination; see <http://psychtoolbox.org/docs/Screen-BlendFunction> [↗](#)). To emulate the functionality of this presentation technique (see <https://github>

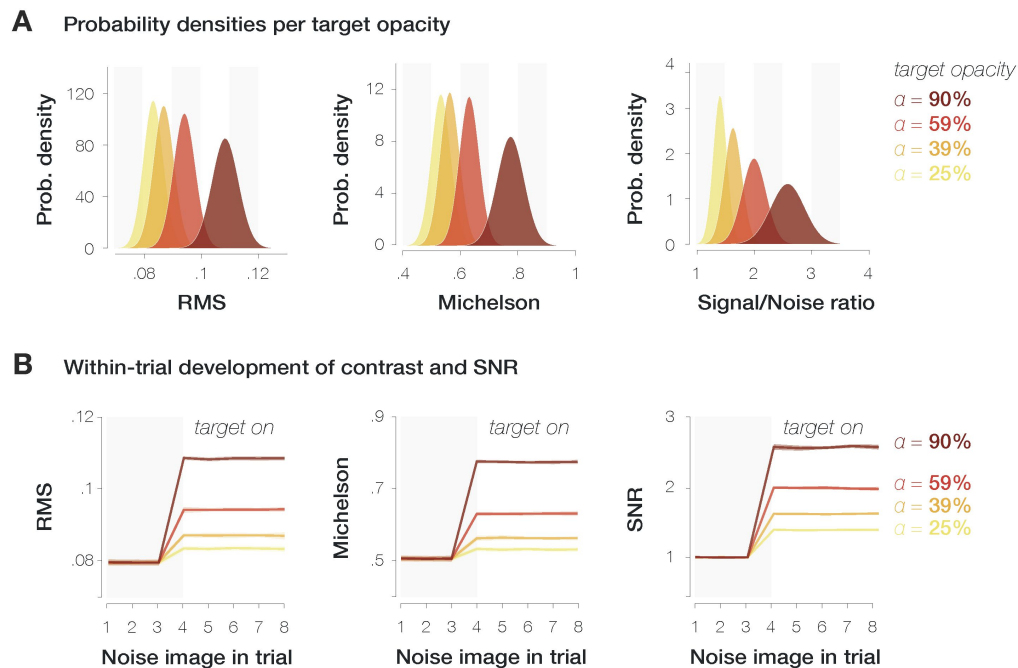


Figure 5.

Variation of RMS contrast (first column), Michelson contrast (second column) and SNR (third column) within the saccade target region.

A. Probability density distributions per measure and target opacity (yellow to dark red shadings). **B.** Mean and standard deviation of contrast and SNR separately for each noise image presented during the saccade preparation period. The target was presented from the 4th noise image on.

[.com/Psychtoolbox-3/Psychtoolbox-3/blob/master/psychtoolbox/](https://www.psychtoolbox.org/psychtoolbox-3/blob/master/psychtoolbox/)

PsychDemos/ContrastModulatedNoiseTheElegantStyleDemo.m; all links retrieved on 2/9/23), we determined the luminance sum of the background noise and the orientation-filtered target patch multiplied with the respective opacity:

$$\text{luminanceCombined} = \text{luminanceBackground} + \text{opacity} * (\text{luminanceFiltered} - \text{luminanceBackground}) * \text{cosineMask}$$

where ‘luminanceCombined’ denotes the resulting luminance values within the target region, ‘luminanceBackground’ denotes the luminance values within a square of 3 dva width around the center of the target location, ‘opacity’ denotes the manipulated target opacity (25-90%), ‘luminanceFiltered’ denotes the luminance values of the orientation-filtered target patch (3 dva width) and ‘cosineMask’ denotes the raised cosine mask with which the filtered patch was overlayed (3 dva diameter). We then calculated the root mean square (RMS, see [Peli, 1990](#)) and Michelson contrast ([Michelson, 1927](#)) of the resulting luminance sum for every trial. To compute the SNR of the target orientation, we determined the energy of various SF and orientation combinations in the combined luminance patch (see **Data analysis**). We then divided the mean filter energy within a region of $\pm 15^\circ$ around the target orientation by the mean filter energy for all remaining orientations. An increase in target opacity from 25% to 90% corresponded to an increase in RMS contrast from .08 to .11, an increase in Michelson contrast from .53 to .78, and an increase in the SNR of the filtered orientation from 1.40 to 2.58 (**Figure 5**).

Data analysis

Online trial abortion criteria as well as trial exclusions based on offline gaze analyses are identical to our previous study ([Kroell & Rolfs, 2022](#)). Unlike in our previous investigation, we intended to remove the saccade target during saccadic flight on all trials. Across observers, 0.24% ($\text{min} = 0\%$; $\text{max} = 1.1\%$; $\text{std} = 0.40\%$) of trials were excluded due to the saccade target still being visible after saccade landing.

To test the influence of target opacity on the difference between congruent and incongruent HRs and FARs, we performed linear mixed-effects models in which we described the variance in both differences ($\text{HR}_{\text{cong}} - \text{HR}_{\text{incong}}$; $\text{FAR}_{\text{cong}} - \text{FAR}_{\text{incong}}$) across all time points with a fixed effect of target opacity and a random intercept for observer:

$$\text{cong-incong} \sim \text{opacity} + (1 \mid \text{observer})$$

section, this model outperformed the simplest model including a fixed effect of target opacity only (1) and a more complex one involving a random intercept and random slope for observer (2):

$$(1) \text{ cong-incong} \sim \text{opacity}$$

$$(2) \text{ cong-incong} \sim \text{opacity} + (\text{opacity} \mid \text{observer})$$

Model fitting was performed with the Matlab function fitlme (Matlab 2020b, Mathworks, Natick, MA, USA).

All remaining tests of statistical significance relied on bootstrapping: Within each observer, we determined the means in the to-be compared conditions and computed the difference between those means. Across observers, we drew 10,000 random samples from these differences (with replacement). Reported p -values correspond to the proportion of differences smaller than or equal to zero. We considered p -values ≤ 0.05 significant.

Furthermore, to relate the difference between congruent and incongruent HRs to the difference in FARs, we z-standardized HR and FAR slopes before computing the Pearson correlation coefficient.

Lastly, we used a three-step approach to calculate the reverse correlation index plotted in **Figure 2C** (see **Figure 6**). First, we determined all noise images that had been visible during saccade preparation on trials in which observers generated a congruent or incongruent FA. We then described the average visual properties of these noise images within a 3 dva diameter circular region around the screen center using a set of Gabor filters with varying SF*orientation properties [SFs from .67 to 2.0 cycles per degree (cpd) in 20 equal steps of .07 cpd; orientations from -90° to 90° in 13 equal steps of 15° ; see Movellan, 2002; Wyart, Nobre, & Summerfield, 2012; Li, Barbot, & Carrasco, 2016; Schweitzer and Rolfs, 2020, 2021]. Second, we subtracted the filter energies on incongruent FA trials from filter energies on congruent FA trials on an individual-observer level. Third, we selected a 30° orientation window around the non-target orientation (i.e., -45°) and subtracted the mean filter energy in that window from the mean filter energy in a 30° orientation window around the target orientation (i.e., 45°). The higher the resulting value, the more congruent and incongruent FAs were based on signal, i.e., on incidental orientation information on screen. Note that we combined the lower two and the higher two target opacities to increase the number of trials and obtain reliable reverse-correlation measures. The resulting index is based on $m = 2,461$ ($std = 583$) and $m = 1,952$ ($std = 629$) trials per observer for the lower two and higher two opacities, respectively.

Supplementary Materials

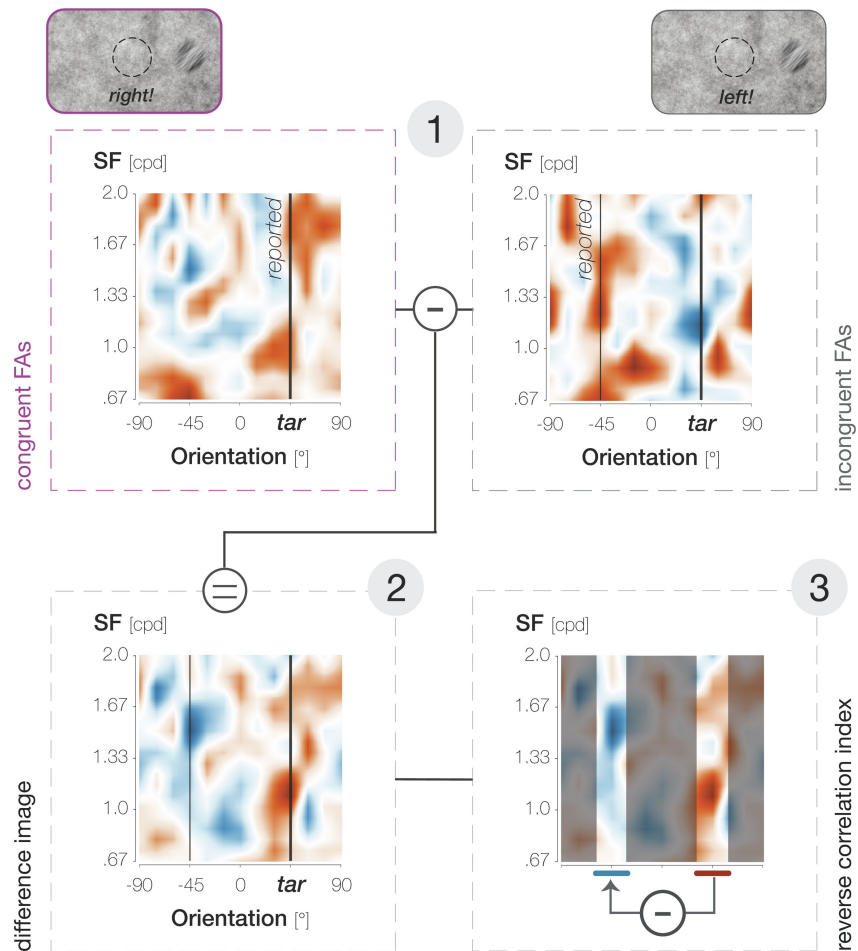


Figure 6.

Calculation of the reverse correlation index plotted in Figure 2C, illustrated for the 59% target opacity condition.

In step 1, we identified the average properties (SF*orientation) of the foveal noise window on congruent (purple outlines and font) and incongruent (gray outlines and font) FA trials. In step 2, we determined the difference between them (congruent-incongruent). In step 3, we subtracted filter energies around the non-target orientation (-45°) from filter energies around the target orientation (45°).

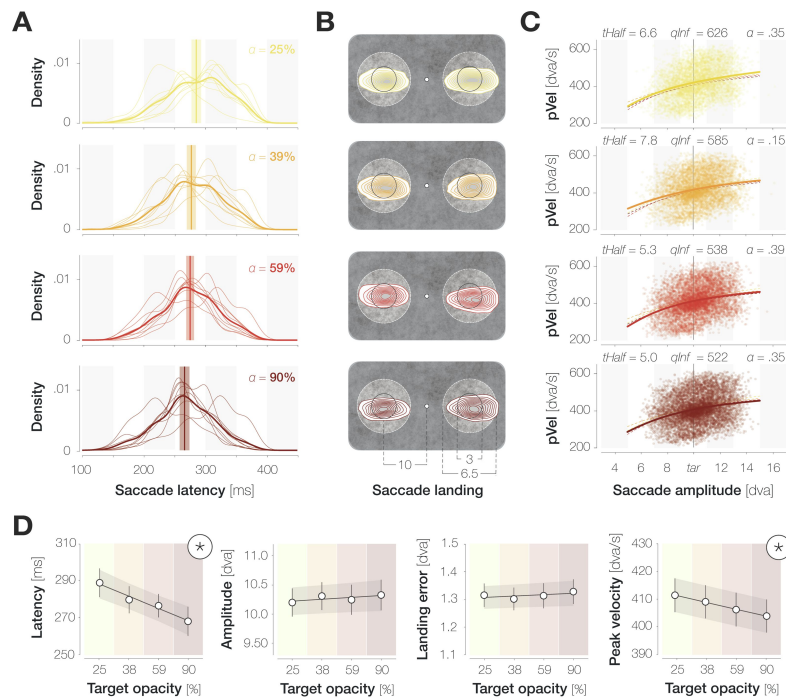


Figure S1.

The influence of target opacity on saccade metrics.

A. Probability density distributions of saccade latencies for different, increasing target opacities (from top to bottom; see **Figure 3A**). Distributions with thin and thick lines represent individual-observer and mean probability densities, respectively. Vertical lines and shaded regions represent median latencies and standard errors. **B.** Bivariate Gaussian kernel densities of saccade landing coordinates separately for leftwards and rightwards saccades. The distance between the fixation and target locations was reduced for illustration purposes (see legend). **C.** Main sequences defined as the relation between saccade amplitudes and peak velocities. Dots symbolize individual trials ($n \sim 29,000$). Fitted lines represent the average of logistic function fits to individual-observer data (Conder, 2023). The mean parameters of each fit are provided above the respective panel ('tHalf': symmetric inflection point; 'qInf': horizontal asymptote; α : decay constant). **D.** Summary plots for saccade latency, amplitude, landing error and peak velocity. Dots represent median (latency) and mean (amplitude, error, velocity) values across observers, and error bars represent the respective standard errors. Black lines and shaded error bands represent the mean of linear fits to individual-observer data and their standard errors, respectively. Asterisks highlight slopes that are significantly different from zero (determined via bootstrapping, $n = 9$ observers, $p < .05$).

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Reviewer #1 (Public review):

Summary:

This study examines to what extent this phenomenon varies based on the visibility of the saccade target. Visibility is defined as the contrast level of the target with respect to the noise background, and it is related to the signal-to-noise ratio of the target. A more visible target facilitates the oculomotor behavior planning and execution, however, as speculated by the authors, it can also benefit foveal prediction even if the foveal stimulus visibility is maintained constant. Remarkably, the authors show that presenting a highly visible saccade target is beneficial for foveal vision as detection of stimuli with an orientation similar to that of the saccade target is improved, the lower is the saccade target visibility, the less prominent is this effect.

Strengths:

The results are convincing and the research methodology is technically sound.

Weaknesses:

It is still unclear why the pre-saccadic enhancement would oscillate for targets with higher opacity levels, and what would be the benefit of this oscillatory pattern. The authors do not speculate too much on this and loosely relate it to feedback processes, which are characterized by neural oscillations in a similar range.

<https://doi.org/10.7554/eLife.91236.2.sa2>

Reviewer #2 (Public review):

Summary:

In this manuscript, the authors ran a dual task. Subjects monitored a peripheral location for a target onset (to generate a saccade to), and they also monitored a foveal location for a foveal probe. The foveal probe could be congruent or incongruent with the orientation of the peripheral target. In this study, the authors manipulated the conspicuity of the peripheral target, and they saw changes in performance in the foveal task. However, the changes were somewhat counterintuitive.

Strengths:

The authors use solid analysis methods and careful experimental design.

Comments on revisions:

The authors have addressed my previous comments.

One minor thing is that I am confused by their assertion that there was no smoothing in the manuscript (other than the newly added time course analysis). Figure 3A and Figure 6 seem to have smoothing to me.

Another minor comment is related to the comment of Reviewer 1 about oscillations. Another possible reason for what looks like oscillations is saccadic inhibition. when the foveal probe appears, it can reset the saccade generation process. when aligned to saccade onset, this appears like a characteristic change in different parameters that is time-locked to saccade onset (about a 100 ms earlier). So, maybe the apparent oscillation is a manifestation of such resetting and it's not really an oscillation. so, I agree with Reviewer 1 about removing the oscillation sentence from the abstract.

<https://doi.org/10.7554/eLife.91236.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Reviews):

Summary:

This study examines to what extent this phenomenon varies based on the visibility of the saccade target. Visibility is defined as the contrast level of the target with respect to the noise background, and it is related to the signal-to-noise ratio of the target. A more visible target facilitates the oculomotor behavior planning and execution, however, as speculated by the authors, it can also benefit foveal prediction even if the foveal stimulus visibility is maintained constant. Remarkably, the authors show that presenting a highly visible saccade target is beneficial for foveal vision as the detection of stimuli with an orientation similar to that of the saccade target is improved, the lower the saccade target visibility, the less prominent the effect.

Strengths:

The results are convincing and the research methodology is technically sound.

Weaknesses:

Discussion on how this phenomenon may unfold in natural viewing conditions when the foveal and saccade target stimuli are complex and are constituted by different visual properties is lacking. Some speculations regarding feedforward vs feedback neural processing involved in the phenomenon and the speed of the feedforward signal in relation to the visibility of the target, are not well justified and not clearly supported by the data.

We thank the reviewer for their comment. In general, we tried to address conceptual points only briefly in this Research Advance if we had discussed them in depth in our main article which this advance will be linked to (Kroell & Rolfs, 2022: <https://elifesciences.org/articles/78106>). However, the reviews showed us that this rendered our theoretical reasoning in the current manuscript appear incomplete. In the revised Discussion section, we have elaborated on several conceptual questions. In particular, we expand on the transferability of our findings to natural viewing conditions:

“Foveal prediction in natural visual environments

As noted above, human observers typically move their eyes towards the most conspicuous objects in their environment (‘t Hart, Schmidt, Roth, & Einhäuser, 2013). Foveal prediction seems to benefit from this strategy as the strength of the predicted signal increases with the conspicuity of the eye movement target. Nonetheless, natural visual environments as well as naturalistic viewing behavior pose several challenges for the foveal prediction mechanism (see Kroell & Rolfs, 2022, for an initial discussion).

First, naturalistic saccade target stimuli will likely exhibit complex shapes and, more often than not, will include feature conjunctions rather than isolated features. Previous findings suggest that the foveal feedback mechanism is capable of operating at this level of complexity: High-level peripheral information such as the category of novel, rendered objects (Williams et al., 2008) has been successfully decoded from activation in foveal retinotopic cortex. If, indeed, temporal objectspecific areas such as area TE send feedback, the foveal prediction mechanism may even be specialized for the transfer of complex visual properties.

Second, foveal input will often be of high contrast in natural visual environments. If fed-back predictive signals can influence foveal perception in the presence of high-contrast feedforward input remains to be established. In our main investigation (Kroell & Rolfs, 2022; Figure 2B) as well as in previous studies (Hanning & Deubel, 2022b), pre-saccadic foveal detection performance decreased markedly in the course of saccade preparation, presumably because visuospatial attention gradually shifted towards the saccade target and away from the foveal location. This presaccadic decrease in foveal sensitivity may boost the relative weight of fed-back signals by attenuating the conspicuity of high-contrast feedforward input. In other words, the strength of feedforward input to the fovea is reduced gradually across saccade preparation. At the same time, the strength of the fed-back predictive signal should profit from the high contrast of naturalistic saccade targets.

Third, while foveal and peripheral information was congruent on 50% of all ‘probe present’ trials in our investigation, peripheral and foveal features will often be weakly correlated or even uncorrelated in natural environments (see Samonds, Geisler, & Priebe, 2018). Again, the presaccadic attenuation of foveal feedforward processing may allow fed-back peripheral signals to influence perception even if they are uncorrelated with foveal information. Moreover, in piloting variations of our paradigm, we observed that the subjective impression of perceiving the saccade target at the pre-saccadic foveal location is most pronounced if the foveal noise region is replaced with a black Gaussian blob at certain time points before saccade onset (unpublished phenomenological accounts). In consequence, fed-back signals do not seem to require correlated feedforward input to influence perception. Quantitative evidence, however, remains to be established.

Lastly, pre-saccadic foveal input is likely less relevant during natural viewing behavior than it is in our task. It is possible that this task-induced prioritization of the foveal location facilitated the emergence of congruency effects. In a previous experiment (Kroell & Rolfs, 2022; Figure 1D), however, the perceptual probe could appear anywhere on a horizontal axis of 9 dva length around the fixation location. Despite this spatial unpredictability, congruency effects peaked at the presaccadic foveal location, even after peripheral baseline performances had been raised to a foveal level through an adaptive increase in probe opacity. On a similar note, the orientation of the saccade target is irrelevant to the behavioral task in our design, mirroring naturalistic situations: The eye movement can be planned and executed based on local contrast variations alone, and observers are never required to report on the orientation of the peripheral target stimulus. Ultimately, however, an influence of task demands on visual processing can only be fully excluded through techniques that provide a direct readout of perceptual contents without requiring overt responses. In psychophysical investigations, a prediction of saccade target motion may be read out from observers’ eye velocities (Kroell, Mitchell, & Rolfs, 2023; Kwon, Rolfs, & Mitchell, 2019). In electroencephalographic (EEG) and electrophysiological studies, foveal predictions should

manifest in early visually evoked potentials (e.g., Creel, 2019) and increased firing rates of featureselective foveal neurons in early visual areas, respectively. In conclusion, previous findings (Williams et al., 2008), the assumed properties of the neuronal feedback mechanism (Williams et al., 2008; Bullier, 2001) and characteristics of our current and previous experimental paradigms collectively suggest that foveal feature predictions are likely to transfer to naturalistic environments and viewing situations. Experimental evidence remains to be established.”

We have furthermore modified the Abstract to emphasize the connection of the current manuscript to the main article.

With respect to the reviewer’s point that “speculations regarding feedforward vs feedback neural processing involved in the phenomenon and the speed of the feedforward signal in relation to the visibility of the target, are not well justified”:

Again, we understand that we should have elaborated on our theoretical reasoning in this Research Advance. The assumption that our initial findings rely on neuronal feedback to foveal retinotopic cortex is derived from Williams et al.’s (2008) seminal findings: In an fMRI study, the category of peripherally presented objects could be decoded from voxels in foveal retinotopic cortex, suggesting that peripheral visual information was available to neurons with strictly foveal receptive fields. We extended these findings to saccade preparation, suggesting that feedback from higher-order, non-retinotopically organized visual areas may transmit information without the requirement of efference copies (see Kroell, 2023; Dissertation; <https://doi.org/10.18452/27204>, pp. 54-59): Irrespective of the vector of the upcoming saccade, the features of the attended saccade target would invariably be relayed to foveal retinotopic cortex. Ultimately, only anatomical and functional studies in non-human primates can conclusively establish the role of feedback connections in the observed foveal prediction effects. At present, however, this parsimonious model could account for all of our current and previous findings, that is, a temporally, spatially and feature-specific anticipation of saccade target properties in the presaccadic center of gaze. Nonetheless, we are open to considering any other mechanism that may account for our findings, and have integrated the explanation provided by the reviewer into the paragraph on potential thalamic mechanisms (see the reviewer’s Major Point 1).

Concerning the point that the “some speculations regarding feedforward vs feedback neural processing [...] and the speed of the feedforward signal in relation to the visibility of the target are not well justified and not clearly supported by the data”:

Theoretical considerations on the impact of peripheral target contrast on feedforward processing speed were a main motivation for the current study. We apologize if our theoretical reasoning was incomplete and have added additional references and elaborations to the Introduction:

“In particular, neuronal response latencies decrease systematically as the contrast of visual input increases. While this phenomenon is reliably observed at varying stages of the visual processing hierarchy—such as the lateral geniculate nucleus (Lee, Elepfandt, & Virsu, 1981b), primary visual cortex (e.g., Albrecht, 1995; Carandini & Heeger, 1994; Carandini, Heeger, & Movshon, 1997; Carandini, Heeger, & Senn, 2002), and anterior superior temporal sulcus (STSa; Oram, Xiao, Dritschel, & Payne, 2002; van Rossum, van der Meer, Xiao, & Oram, 2008)—influences of contrast on neuronal response latency are particularly pronounced in higher-order visual areas: A doubling of stimulus contrast has been shown to decrease the latency of V1 neurons by 8 ms, compared to a reduction of 33 ms in area STSa (Oram et al., 2002; van Rossum et al., 2008). Assuming that the peripheral target is processed in a bottom-up fashion until it reaches higher-order object processing areas, the time point at which peripheral signals are available for feedback should be dictated by the temporal dynamics of visual feedforward processing.”

Concerning the interpretation of the observed time courses, and regarding the reviewer's Major points 3 & 6, we substantially revised the Results and Discussion section. In brief, we deemphasized the claim/interpretation of faster enhancement with increasing target opacity and instead focus on describing the oscillatory pattern mentioned by the reviewer. We provide a more temporally resolved pre-saccadic time course using a moving-window analysis and discuss all suggested and further alternative explanations (i.e., saccade-locked perceptual or attentional oscillations, longer signal accumulation intervals for low-contrast information, oscillatory nature of feedback signaling). Details and full revised paragraphs are provided in the response to this reviewer's Major points 3 & 6.

Unfortunately, there is no line numbering in the manuscript version I downloaded so I cannot refer to the specific lines of text here.

We apologize for the inconvenience and have added line numbers.

Major:

(1) The authors speculate that the phenomenon of pre-saccadic foveal prediction arises from feedback connections from higher-order visual areas, which relay relevant saccade target features to the foveal retinotopic cortex. These feedback signals are then presumably combined with feedforward foveal input to the early visual cortex and facilitate the detection of target-congruent features at the center of gaze. This interpretation is sensible, however, it may not be the only plausible scenario. The thalamus receives copies of feedforward and feedback connections between all visual areas and is a likely candidate hub for combining information across visual space. In this latter case, the phenomenon of pre-saccadic foveal prediction may not arise from feedback from higher-order visual areas, but rather from a combination of signals occurring at the level of the thalamus. The authors should either acknowledge this possibility and the fact that this phenomenon is not necessarily the result of a feedback loop, or they should explain their rationale for excluding this scenario.

We thank the reviewer for their highly thoughtful suggestion, and for alerting us to relevant literature. We have added the following paragraph to the Discussion section. In brief, we discuss the thalamic pulvinar as either an intermediate modulatory region or as the final receiver of the fed-back signal. Yet, we assume that—to solve the combinatorial issue associated with a transfer of feature information before saccades with any possible direction and amplitude—the contribution of non-retinotopic, higherorder object processing areas is likely required.

“Neural implementation of foveal prediction

Based on the body of our findings as well as previous literature, we suggested a parsimonious feedback mechanism to underly the observed effects: the preparation of a saccadic eye movement, and the concomitant shift of pre-saccadic attention (e.g., Kowler, Anderson, Doshier, & Blaser, 1995; Deubel & Schneider, 1996), selects the peripheral target stimulus among competing information. Higher-order visual areas feed selected feature input back to early retinotopic areas—specifically, to neurons with foveal receptive fields. Fed-back feature information combines with congruent, foveal feedforward input, resulting in the enhancement effects we observe. Especially in the context of active vision, this feedback mechanism is appealing as it resolves a combinatorial issue associated with feature-specific information transfer before saccades. Consider a simplified case in which, right before a saccadic eye movement, the activation of a feature-selective neuron that encodes a certain retinal location is transferred to a neuron within the same brain area that will encode said retinal location after saccade landing. For this mechanism to function for any possible

saccade direction and amplitude, most neurons would need to be connected to most other neurons (or, in a simplified version, to neurons with foveal receptive fields) in a given brain area. Assuming an information transmission via feedback rather than horizontal connections significantly reduces this dimensionality: Higher-order visual areas that encode object properties (largely) detached from retinotopic or spatiotopic reference frames selectively transfer feature information to neurons with foveal receptive fields, irrespective of the vector of the upcoming saccade. This parsimonious mechanism would have shortcomings. In particular, foveal feedback should become less effective during saccade sequences where several peripheral targets are simultaneously attended. Feature information at both attended target locations may be fed back in temporal succession or weighted and erroneously combined into a single fed-back signal. In most cases, however, foveal feedback may reasonably achieve what established transsaccadic mechanisms struggle to explain: An anticipation of the features of a single saccade target—which typically constitutes the currently most relevant object in the visual field—in foveal vision.

While direct feedback connections from higher-order to early visual areas would constitute the most straightforward implementation, it is conceivable that feedback signals are relayed through and modulated by subcortical areas. In particular, the thalamic pulvinar has been identified as a connection hub for visual processing that receives copies of feedforward and feedback connections from different visual areas and may even combine information across visual space (Cortes, Ladret, Abbas-Farishta, & Casanova, 2024). In the case of foveal prediction, thalamic neurons may receive fed-back signals from higher-order areas and enhance those signals before passing them on to cortical neurons with foveal receptive fields. Perhaps, a modification of foveal activation within the thalamic pulvinar itself is sufficient to influence perception. To the best of our understanding, however, the fed-back signal must originate in non-retinotopic, higher-order object processing areas to reduce the number of necessary neuronal connections.”

(2) The results presented are very compelling. I wonder to which extent they generalize to situations in which the foveal input and the peripheral input are more heterogeneous (e.g., faces or complex objects composed of many different features, orientations, and other visual properties). I think the current research raises a number of interesting questions. In general, it would be important for the readers to elaborate more on how the mechanism of pre-saccadic foveal prediction may play out in normal viewing conditions or in conditions in which the foveal input is completely irrelevant to the task.

We agree and have reiterated this point in the current manuscript (see our first reply to “Weaknesses”). We also explicitly refer to Kroell & Rolfs (2022) for an extensive initial discussion of this question.

(3) On page 10 the authors state that their data suggest that foveal enhancement emerges in earlier stages of saccade preparation as target opacity increases. However, this is not clear from the figures, when performance is locked to saccade onset (Fig 3 C), for the highest opacity targets performance seems to oscillate, however, the authors do not comment on that. There is literature showing how saccades can reset perceptual oscillations, and maybe what is observed here is just a stronger performance oscillation when the saccade target is more visible. Why would performance drop systematically 75 ms before saccade onset and then increase again 25 ms before the onset? Can the authors elaborate more on this?

In response to this comment, we inspected the pre-saccadic time course of enhancement effects in a more temporally resolved fashion and, indeed, observed pronounced oscillations for the two higher target opacity conditions (see Results):

“Especially at higher target opacities, the temporal development of foveal enhancement appears to exhibit an oscillatory pattern. To inspect this incidental observation in a more temporally resolved fashion, we determined mean enhancement values in a boxcar window of 50 ms duration sliding along all saccade-locked probe offset time points (step size = 10 ms; x-axis values in Figure 4 indicate the latest time point in a certain window). We then fitted 6th order polynomials (with no constraints on parameters) to the resulting time courses and compared the fitted values against zero using bootstrapping (see Methods). The average foveal enhancement across target opacities reached significance starting 115 ms before saccade onset (gray curve in Figure 4; all p s < .046). For every individual target opacity condition, we observed significant enhancement immediately before saccade onset, although only very briefly for the lowest opacity (-2–0 ms for 25%; -39–0 ms for 39%, -106–0 ms for 59% & -13–0 ms for 90%; all p s < .050; yellow to dark red curves in Figure 4). Especially for the higher two target opacities, we observed a local maximum preceding eye movement onset by approximately 80 ms. Interestingly, assuming a peak in enhancement in approximately 80 ms intervals (i.e., at x-axis values of -80 and 0 ms in Figure 4) would correspond to an oscillation frequency of 12.5 Hz. In contrast to rapid feedforward processing, feedback signaling is associated with neural oscillations in the alpha and beta range (i.e., between 7 and 30 Hz; Bastos et al., 2015; Jensen, Bonnefond, Marshall, & Tiesinga, 2015; van Kerkoerle et al., 2015).”

We had observed an oscillatory pattern in multiple previous investigations, and in both Hit Rates to foveal orientation content and reflexive gaze velocities in response to peripheral motion information. So far, we have been unsure how to explain it. The literature on thalamic visual processing mentioned by the reviewer alerted us to the oscillatory nature of feedback signaling itself. Interestingly, the temporal frequency range of feedback oscillations includes the frequency of ~12.5 Hz observed in our data. We have included this and alternative explanations in the Discussion section (see below). Throughout, we highlight that we are aware that our analysis approach is purely descriptive and that the potential explanations we give are speculative.

“Moreover, foveal congruency effects appear to exhibit an oscillatory pattern, with peaks in a medium saccade preparation stage (~80 ms before the eye movement) and immediately before saccade onset. We have noticed this pattern in several investigations with substantially different visual stimuli and behavioral readouts. For instance, using a full-screen dot motion paradigm, we observed a pre-saccadic, small-gain ocular following response to coherent motion in the saccade target region (Kroell, Rolfs, & Mitchell, 2023, conference abstract; Kroell, 2023, dissertation). Predictive ocular following first reached significance ~125 ms before the eye movement, then decreased and subsequently ramped up again ~25 ms before saccade onset. Several explanatory mechanisms appear conceivable. Unlike rapid feedforward processing, feedback propagation has been shown to follow an oscillatory rhythm in the alpha and beta range, that is, between 7 and 30 Hz (Bastos et al., 2015; Jensen, Bonnefond, Marshall, & Tiesinga, 2015; van Kerkoerle, et al., 2015). In our case, it is possible that the object-processing areas that send feedback to retinotopic visual cortex do so at a temporal frequency of ~12.5 Hz. At higher stimulus contrasts, feedforward signals may be fed back instantaneously and without the need for signal accumulation in feedback-generating areas. The resulting perceptual time courses may reflect innate temporal feedback properties most veridically. Alternatively, the initial enhancement peak may be related to the sudden onset of the saccade target stimulus and not to movement preparation itself. In this case, the initial peak should become particularly apparent if enhancement is aligned to the onset of the target stimulus. Yet, Figure 3 and Figure 4 suggest more prominent oscillations in saccade-locked time courses. In accordance with this, perceptual and attentional processes have been shown to exhibit oscillatory modulations that are phase-locked to action onset (e.g., Tomassini, Spinelli, Jacono, Sandini, & Morrone, 2015; Hogendoorn, 2016; Wutz, Muschter, van Koningsbruggen, Weisz, & Melcher, 2016; Benedetto & Morrone, 2017; Tomassini, Ambrogioni, Medendorp, & Maris, 2017; Benedetto, Morrone &

Tomassini, 2019). Whether the oscillatory pattern of foveal enhancement, as well as its increased prominence at higher target contrasts, relies on innate temporal properties of feedback signaling, signal accumulation, saccade-locked oscillatory modulations of feedforward processing or attention, or a combination of these factors, one conclusion remains: task-induced cognitive influences suggested to underlie the considerable variability in temporal characteristics of foveal feedback during passive fixation (e.g., Fan et al., 2016; Weldon et al., 2016; 2020) are not the only possible explanation. Low-level target properties such as its luminance contrast modulate the resulting time course and should be equally considered, at least in our paradigm.”

In the revised Abstract, we removed our claim on an earlier emergence of enhancement at higher opacities and have added this summary instead:

“Second, the time course of foveal enhancement appeared to show an oscillatory pattern that was particularly pronounced at higher target opacities. Interestingly, the temporal frequency of these oscillations corresponded to the frequency range typically associated with neural feedback signaling.”

(4) What was the average difference in latency between short and long latencies? It would be good to report it in the main text.

We apologize for the oversight. The difference was 61 ms, with latencies of $md = 247 \pm 18$ ms for short- and $md = 308 \pm 18$ ms for long-latency saccades. We have added this information to the main text.

(5) From the saccade latency graphs in Figure S1 it seems there is some variability in the latency of saccades across subjects, I wonder if there is a correlation between saccade latency and the magnitude of the foveal prediction effect across subjects.

We had inspected a connection between saccade latency and congruency in our first investigation (Kroell & Rolfs, 2022; not reported) and observed that participants with lower latencies tended to show more enhancement, albeit non-significantly. Likewise, we observed a non-significant negative correlation between the median saccade latency and the mean foveal prediction effect (across opacities and time points) in the current investigation, $r = -0.22$, $p = .572$. While our study involved a small number of observers ($n = 9$), the analysis approach illustrated in Figure 2 A-C instead makes use of the large number of trials collected per participant (mean $n = 2841$ trials per observer) and demonstrates a reliable influence of saccade latency on an individual-observer level.

(6) Page 14, the authors state that their findings suggest that the feedforward processing of the peripheral saccade target is accelerated when it is presented at high contrast. I find this a bit too speculative, both in terms of assuming that there is a feedforward vs a feedback process (see my point 1) and in terms of speculating that the feedforward process is accelerated as I do not see a clear hint of this in the data (see my point 3) and it is a bit of a stretch to speculate on delays or accelerations of neural processing. It is possible that the feedforward signal is always delivered at the same speed but it is weaker in one case and the effect needs more time to build up.

We fully agree and hope to have addressed the reviewer’s arguments in the sections preceding this point. We included the reviewer’s last sentence in the Discussion section as well:

“Alternatively, or in addition, it is conceivable that weaker feedforward signals require a longer accumulation interval before the feedback process can be initiated.”

(1) I think the description of the linear mixed-effects model can go in the supplemental methods, if possible, and its results can be briefly mentioned in the text.

In previous work, we have been asked to move linear mixed-effects model descriptions from supplemental to main method (or even results) sections for clarity. We have followed this suggestion ever since and, due to the relevance of the models for the interpretation of the presented results, would like to keep their description in the methods section.

(2) This is just a minor point, but I would suggest using a different word instead of opacity (maybe visibility?).

We had gone back and forth on this. We decided to use the term ‘conspicuity’ when we discuss our findings conceptually and the term ‘opacity’ when we refer to the experimental manipulation (since we directly manipulate the transparency, i.e., 1-opacity, of the target patch against the background). To compute the slopes in Figures 2 and 5, we ordered observers’ performances by the linearly spaced opacity conditions. Since the term ‘opacity’ is closest to both the experimental manipulation and the variable entered into analysis, we would like to adhere to this terminology. However, we have added an explicit note to the end of our introduction to avoid confusion:

“Throughout the paper, we use the term ‘opacity’ when we refer to the experimental manipulation (that is, a variation of the transparency, i.e., 1-opacity of the target patch against the background noise) and the term ‘conspicuity’ when we discuss our findings conceptually.”

Reviewer #2 (Public Review):

Summary:

In this manuscript, the authors ran a dual task. Subjects monitored a peripheral location for a target onset (to generate a saccade to), and they also monitored a foveal location for a foveal probe. The foveal probe could be congruent or incongruent with the orientation of the peripheral target. In this study, the authors manipulated the conspicuity of the peripheral target, and they saw changes in performance in the foveal task. However, the changes were somewhat counterintuitive.

Strengths:

The authors use solid analysis methods and careful experimental design.

Weaknesses:

I have some issues with the interpretation of the results, as explained below. In general, I feel that a lot of effects are being explained by attention and target-probe onset asynchrony etc, but this seems to be against the idea put forth by the authors of "foveal prediction for visual continuity across saccades". Why would foveal prediction be so dependent on such other processes? This needs to be better clarified and justified.

We address the described weaknesses in the respective sections below. In general, as we point out in response to Reviewer 1 as well, the current submission is a Research Advance article meant to supplement our main article (Kroell & Rolfs, 2022, <https://doi.org/10.7554/eLife.78106>). To comply with the eLife recommendations for Research Advance submissions, we addressed conceptual points only briefly, especially if they had been explained in detail in our main article. To make the nature and format of the current submission as explicit as possible, and to emphasize its connection to our previous work, we refer to the submission format in our abstract and introduction now.

Specifics:

The explanation of decreased hit rates with increased peripheral target opacity is not convincing. The authors suggest that higher contrast stimuli in the periphery attract attention. But, then, why are the foveal results occurring earlier (as per the later descriptions in the manuscript)? And, more importantly, why would foveal prediction need to be weaker with stronger pre-saccadic attention to the periphery? What is the function of foveal prediction? What of the other interpretation that could be invoked in general for this type of task used by the authors: that the dual task is challenging and that subjects somehow misattribute what they saw in the peripheral task when planning the saccade. i.e. foveal hit rates are misperceptions of the peripheral target. When the peripheral target is easier to see, then the foveal hit rate drops.

We will address these comments one by one:

The authors suggest that higher contrast stimuli in the periphery attract attention. But, then, why are the foveal results occurring earlier (as per the later descriptions in the manuscript)?

We consider these observations to rely on separate processes. Already in the main publication (Kroell & Rolfs, 2022), we had observed a continuous decrease of target-congruent *and* target-incongruent foveal Hit Rates (HRs) during saccade preparation, and suggested that this decrease (similarly observed in Hanning & Deubel, 2022b) is likely caused by the pre-saccadic shift of visuospatial attention to the target. In other words, as attentional resources shift towards the periphery, foveal detection performance is hampered, irrespective of peripheral and foveal feature (in-)congruency. In the current investigation, we again observed a pronounced pre-saccadic decrease of foveal HRs, irrespective of foveal probe orientation. Our argument that high-contrast peripheral saccade targets attract more attention relies on the clear observation that this decrease becomes more pronounced as the contrast of the saccade target increases. To the best of our judgment and experience with doing the task ourselves, this interpretation appears very conceivable. We explain this rationale in the Abstract and the Results sections of the manuscript (see below).

Our hypotheses and interpretations concerning the time course of foveal prediction refer to the *difference* between target-congruent and target-incongruent foveal HRs (i.e., to predictive foveal feature enhancement). Irrespective of the general, feature-unspecific decrease of foveal detection performances, we had hypothesized that the peripheral target is processed faster if it exhibits a high contrast. This assumption is based on temporal processing properties of many visual neurons that we have expanded on in our revision:

“In particular, neuronal response latencies decrease systematically as the contrast of visual input increases. While this phenomenon is reliably observed at varying stages of the visual processing hierarchy—such as the lateral geniculate nucleus (Lee et al., 1981b), primary visual cortex (e.g., Albrecht, 1995; Carandini et al., 1997, 2002; Carandini and Heeger, 1994), and anterior superior temporal sulcus (STSa; Oram et al., 2002; van Rossum et al., 2008)—influences of contrast on neuronal response latency are particularly pronounced in higher-order visual areas: A doubling of stimulus contrast has been shown to decrease the latency of V1 neurons by 8 ms, compared to a reduction of 33 ms in area STSa (Oram et al., 2002; van Rossum et al., 2008). Assuming that the peripheral target is processed in a bottom-up fashion until it reaches higher-order object processing areas, the time point at which peripheral signals are available for feedback should be dictated by the temporal dynamics of visual feedforward processing.”

Of note, both reviewers asked us to explore the oscillatory nature of the difference between targetcongruent and target-incongruent HRs. We will post our changes in response to the reviewer's remark below.

And, more importantly, why would foveal prediction need to be weaker with stronger pre-saccadic attention to the periphery?

We hope that our previous reply has cleared up that the opposite is true: In general, and irrespective of the feature congruency of target and foveal probe, foveal HRs decrease as target contrast increases. As we have stated in our Abstract and Results, “foveal Hit Rates for target-congruent and incongruent probes decreased as target opacity increased, presumably since attention was increasingly drawn to the target the more salient it became. Crucially, foveal enhancement defined as the difference between congruent and incongruent Hit Rates increased with opacity”. This finding did not appear counterintuitive to us and was, in fact pre-registered as a main hypothesis (see <https://osf.io/wceba>).

We are unsure if this goes beyond the reviewer's concern but we, in fact, speculate in the revised Discussion section as well as in our original eLife article that the overall, feature-unspecific decrease in foveal detection performances may aid feature-specific foveal prediction:

“This pre-saccadic decrease in foveal sensitivity may boost the relative weight of fed-back signals by attenuating the conspicuity of high-contrast feedforward input. In other words, the strength of feedforward input to the fovea is reduced gradually across saccade preparation. At the same time, the strength of the fed-back predictive signal should profit from the high contrast of naturalistic saccade targets.”

What is the function of foveal prediction?

Please refer to the section ‘What is the function of foveal prediction?’ in our main article. We have pasted this paragraph below for the reviewer's convenience.

“What is the function of foveal prediction?

As stated above, previous investigations on foveal feedback required observers to make peripheral discrimination judgments. We, in contrast, did not ask observers to generate a perceptual judgment on the orientation of the saccade target. Instead, detecting the target was necessary to perform the oculomotor task. While the identification of local contrast changes would have sufficed to direct the eye movement, the orientation of the target enhanced foveal processing of congruent orientations. The automatic nature of foveal enhancement showcases that perceptual and oculomotor processing are tightly intertwined in active visual settings: planning an eye movement appears to prioritize the features of its target; commencing the processing of these features before the eye movement is executed may accelerate post- saccadic target identification and ultimately provide a head start for corrective gaze behavior (Deubel et al., 1982; Ohl and Kliegl, 2016; Tian et al., 2013).”

What of the other interpretation that could be invoked in general for this type of task used by the authors: that the dual task is challenging and that subjects somehow misattribute what they saw in the peripheral task when planning the saccade. i.e. foveal hit rates are misperceptions of the peripheral target. When the peripheral target is easier to see, then the foveal hit rate drops.

Alternative explanations in general: In our main article, we ruled out—either through direct experimentation or by considering relevant properties of our findings—the following alternative explanations: i) spatially global feature-based attention to the target orientation,

ii) a multiplicative combination of spatial and feature-based attention, and iii) shifts of decision criterion. While dual tasks (i.e., simultaneous oculomotor planning and perceptual detection) are standard in psychophysical investigations of active vision, we acknowledge the potential influence of an explicit foveal task in the revised manuscript, and in response to both reviewers:

“Lastly, pre-saccadic foveal input is likely less relevant during natural viewing behavior than it is in our task. It is possible that this task-induced prioritization of the foveal location facilitated the emergence of congruency effects. In a previous experiment (Kroell & Rolfs, 2022; Figure 2D), the perceptual probe could appear anywhere on a horizontal axis of 9 dva length around the screen center. Despite this spatial unpredictability, however, congruency effects peaked at the pre-saccadic foveal location, even after peripheral baseline performances had been raised to a foveal level through an adaptive increase in probe opacity. Ultimately, an influence of task demands on visual processing can only be fully excluded through techniques that provide a direct readout of perceptual contents without requiring keyboard responses. In psychophysical investigations, a prediction of saccade target motion may be read out from observers’ eye velocities (Kroell, Mitchell, & Rolfs, 2023; Kwon, Rolfs, & Mitchell, 2019). In electroencephalographic (EEG) and neurophysiological studies, foveal predictions should manifest in early visual evoked potentials (e.g., Creel, 2019) and increased firing rates of feature-selective foveal neurons in early visual areas, respectively.”

Difficulty of the task: Concerning the perceptual detection task, every experimental session was preceded by an adaptive staircase procedure that adjusted the transparency of the foveal probe—and, thus, task difficulty—depending on the respective observer’s performance (see Methods for details). Concerning the oculomotor task, observers were able to perform accurate saccades with typical movement latencies for all target opacity conditions (see Results, Supplements & Figure S1). In general, we are unsure how high task difficulty could produce a feature-, temporally and spatially specific enhancement of both filtered and incidental target-congruent foveal orientation information. In fact, a main finding of our current submission is that foveal HRs decrease as the target becomes easier to see and the oculomotor task thus becomes easier to perform.

Perceptual confusion of target and probe stimulus: We observe a specific increase in HRs for foveal probes that exhibit the same orientation as the peripheral saccade target. Just like in our main article, a response is defined as a ‘Hit’ if a foveal probe is presented and the observer generates a ‘present’ judgment. To our understanding, the suggestion that a confusion of target and probe stimuli may account for these effects necessarily implies that this confusion hinges on the congruency between peripheral and foveal feature inputs. In other words, peripheral and foveal signals should be more readily “confused” if they exhibit similar features. We assume that peripheral feature information is fed back to neurons with foveal receptive field and combines with feature-congruent feedforward input. Whether this combination of signals can be described as low-level perceptual “confusion” likely depends on individual linguistic judgments (it would certainly be a novel description of feedback-feedforward interactions). Perhaps a defining difference between the reviewer’s concern and our assumed mechanism is the spatial specificity of the resulting congruency effects. We suggest that only neurons with foveal receptive fields receive feature information via feedback. And indeed, we demonstrate a clear spatial specificity of congruency effects around the pre-saccadic foveal location, even after parafoveal performances had been raised to a foveal level by an adaptive increase in probe opacity (see Kroell & Rolfs, 2022; Figure 2C & Figure 3). In other words, observers’ perception is altered in their pre-saccadic center of gaze while the target is presented peripherally. We struggle to conceive a

scenario in which a confusion of signals should be feature-specific as well as specific to an interaction between peripheral and foveal signals without being meaningful at the same

time. If the reviewer is referring to confusions on the response or decision level, we would like to point them towards the Discussion section ‘Can our findings be explained by established mechanisms other than foveal prediction?’ in our main article. In this paragraph, we provide detailed arguments for a dissociation between our findings and shifts in decision criterion that would exceed the scope of a Research Advance.

When the peripheral target is easier to see, then the foveal hit rate drops.

We agree. Target-congruent and incongruent foveal HRs decreased as the contrast of the probe increased. However, and as we stated in response to the reviewer’s first comment, the difference between target-congruent and target-incongruent foveal HRs (and, thus, foveal enhancement of the target orientation) increased with peripheral target contrast.

The analyses of Fig. 3C appear to be overly convoluted. They also imply an acknowledgment by the authors that target-probe temporal difference matters. Doesn't this already negate the idea that the foveal effects are associated with the saccade generation process itself? If the effect is related to target onset, how is it interpreted as related to a foveal prediction that is associated with the saccade itself?

We indeed conducted analyses that can reveal an influence of target presentation duration at probe onset, the saccade preparation stage at probe offset, as well as a combination of both factors. The fact that target presentation duration may have an influence on foveal prediction would not negate a simultaneous influence of saccade preparation and vice versa. In the main article, we directly investigated the influence of saccade preparation on foveal enhancement by introducing a passive fixation condition (Kroell & Rolfs, 2022; Figure 5). At identical target-probe offset durations, pre-saccadic foveal enhancement was significantly more pronounced and accelerated compared to enhancement during passive fixation. We have added a purely saccade-locked time course (uncorrected by targetprobe interval) to our Results section and to Figure 3 (second row). We still believe that the target-locked, saccade-locked and combined analysis are informative for future investigations and would like to present them all for completeness.

Also, the oscillatory nature of the effect in Fig. 3C for 59% and 90% opacity is quite confusing and not addressed. The authors simply state that enhancement occurs earlier before the saccade for higher contrasts. But, this is not entirely true. The enhancement emerges then disappears and then emerges again leading up to the saccade. Why would foveal prediction do that?

In response to this comment and a suggestion by Reviewer 1, we inspected the pre-saccadic time course of enhancement effects in a more temporally resolved fashion and, indeed, observed pronounced oscillations for the two higher target opacity conditions (see Results):

“Especially at higher target opacities, the temporal development of foveal enhancement appears to exhibit an oscillatory pattern. To inspect this incidental observation in a more temporally resolved fashion, we determined mean enhancement values in a boxcar window of 50 ms duration sliding along all saccade-locked probe offset time points (step size = 10 ms; x-axis values in Figure 4 indicate the latest time point in a certain window). We then fitted 6th order polynomials to the resulting time courses and compared the fitted values against zero using bootstrapping (see Methods). The average foveal enhancement across target opacities reached significance starting 115 ms before saccade onset (gray curve in Figure 4; all $ps < .046$). For every individual target opacity condition, we observed significant enhancement immediately before saccade onset, although only very briefly for the lowest opacity (-2–0 ms for 25%; -39–0 ms for 39%, -106–0 ms for 59% & -13–0 ms for 90%; all $ps < .050$; yellow to dark red curves in Figure 4). Especially for the higher two target opacities, we observed a local

maximum preceding eye movement onset by approximately 80 ms. Interestingly, assuming a peak in enhancement in approximately 80 ms intervals (i.e., at x-axis values of -80 and 0 ms in Figure 4) would correspond to an oscillation frequency of 12.5 Hz. In contrast to rapid feedforward processing, feedback signaling is associated with neural oscillations in the alpha and beta range (i.e., between 7 and 30 Hz; Bastos et al., 2015; Jensen, Bonnefond, Marshall, & Tiesinga, 2015; van Kerkoerle et al., 2015)."

We had observed an oscillatory pattern in multiple previous investigations, and in both Hit Rates to foveal orientation content and reflexive gaze velocities in response to peripheral motion information. So far, we have been unsure how to explain it. The literature on thalamic visual processing mentioned by the reviewer alerted us to the oscillatory nature of feedback signaling itself. Interestingly, the temporal frequency range of feedback oscillations includes the frequency of ~12.5 Hz observed in our data. We have included this and alternative explanations in the Discussion section (see below). We are aware, and acknowledge in the manuscript, that our analysis approach is purely descriptive, and that the potential explanations we give are speculative.

"Moreover, foveal congruency effects appeared to exhibit an oscillatory pattern, with peaks in a medium saccade preparation stage (~80 ms before the eye movement) and immediately before saccade onset. We have noticed this pattern in several investigations with substantially different visual stimuli and behavioral readouts. For instance, using a full-screen dot motion paradigm, we observed a pre-saccadic, small-gain ocular following response to coherent motion in the saccade target region (Kroell, Rolfs, & Mitchell, 2023, conference abstract; Kroell, 2023, dissertation). Predictive ocular following first reached significance ~125 ms before the eye movement, then decreased and subsequently ramped up again ~25 ms before saccade onset. Several explanatory mechanisms appear conceivable. Unlike rapid feedforward processing, feedback propagation has been shown to follow an oscillatory rhythm in the alpha and beta range, that is, between 7 and 30 Hz (Bastos et al., 2015; Jensen, Bonnefond, Marshall, & Tiesinga, 2015; van Kerkoerle, et al., 2015). In our case, it is possible that the object-processing areas that send feedback to retinotopic visual cortex do so at a temporal frequency of ~12.5 Hz. At higher stimulus contrasts, feedforward signals may be fed back instantaneously and without the need for signal accumulation in feedback-generating areas. The resulting perceptual time courses may reflect innate temporal feedback properties most veridically. Alternatively, the initial enhancement peak may be related to the sudden onset of the saccade target stimulus and not to movement preparation itself. In this case, the initial peak should become particularly apparent if enhancement is aligned to the onset of the target stimulus. Yet, Figure 3 and Figure 4 suggest more prominent oscillations in saccade-locked time courses. In accordance with this, perceptual and attention processes have been shown to exhibit oscillatory modulations that are phase-locked to action onset (e.g., Tomassini, Spinelli, Jacono, Sandini, & Morrone, 2015; Hogendoorn, 2016; Wutz, Muschter, van Koningsbruggen, Weisz, & Melcher, 2016; Benedetto & Morrone, 2017; Tomassini, Ambrogioni, Medendorp, & Maris, 2017; Benedetto, Morrone & Tomassini, 2019). Whether the oscillatory pattern of foveal enhancement, as well as its increased prominence at higher target contrasts, relies on innate temporal properties of feedback signaling, signal accumulation, saccade-locked oscillatory modulations of feedforward processing or attention, or a combination of these factors, one conclusion remains: task-induced cognitive influences suggested to underlie the considerable variability in temporal characteristics of foveal feedback during passive fixation (e.g., Fan et al., 2016; Weldon et al., 2016; 2020) are not the only possible explanation. Low-level target properties such as its luminance contrast modulate the resulting time course and should be equally considered, at least in our paradigm."

The interpretation of Fig. 4 is also confusing. Doesn't the longer latency already account for the lapse in attention, such that visual continuity can proceed normally now that the saccade is actually eventually made? In all results, it seems that the effects are all related

to the dual nature of the task and/or attention, rather than to the act of making the saccade itself. Why should visual continuity (when a saccade is actually made, whether with short or long latency) have different "fidelity"? And, isn't this disruptive to the whole idea of visual continuity in the first place?

We are unsure if we grasp the unifying concern behind these remarks. For the reviewer's point on the dual-task nature of our paradigm, please consider our answer above. Perhaps it is important to note that we do not (and would never) claim that foveal prediction is the only mechanism underlying visual continuity. We believe that multiple mechanisms, including but not limited to pre-saccadic shifts of attention, predictive remapping of attention pointers and the perception of intra-saccadic signals interact and jointly contribute to visual continuity. It appears highly conceivable that, like most processes in biological systems, motor and perceptual performances are subject to fluctuations. We argue that saccade latencies as well as the magnitude of foveal prediction constitute read-outs of these variations. We also suggest that those read-outs are innately correlated beyond their common moderator of, perhaps, attentional state; we have previously presented clear evidence for a link between eye movement preparation and foveal prediction (Kroell & Rolfs, 2022; Figure 2). To the best of our judgment, we consider it reasonable that the effectiveness of movement-contingent perceptual processes varies with the effectiveness (in programming or execution) of the very movement motivating them. We present evidence for this assumption in our submission. We would also like to make clear that we do not assume our vision to fail entirely, even if every single well-known mechanism of visual continuity were to break down at once. Upon saccade landing, the visual system receives reliable visual input. Nonetheless, the visual system has undeniably developed mechanisms to optimize this process. We believe foveal prediction to rank among them.

Small question: is it just me or does the data in general seem to be too excessively smoothed?

We did not apply any smoothing to either the analysis or visualization of our data in the initial manuscript.

Every observer completed a large number of trials (mean $n = 2841$ trials per observer; total trial number $> 25,500$), which likely contributes to the clarity of our data. To inspect the oscillatory pattern of enhancement in a more temporally resolved fashion (in response to the reviewer's point above), we applied a moving window analysis in this revision. Due to overlapping window borders, this analysis introduces a certain degree of smoothing. Nonetheless, data patterns are comparable to the time course with only few non-overlapping time bins (Figure 3B; second row). In general, we have described all steps of our analysis routine extensively in the Methods section and will make our data publicly available upon publication of the Reviewed Preprint.

General comment: it is important to include line numbers in manuscripts, to help reviewers point to specific parts of the text when writing their comments. Otherwise, the peer review process is rendered unnecessarily complicated for the reviewers.

We apologize and have added line numbers.

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